

# The Alberta Buck - Identity - Cryptography Example (v1.0)

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This document has two parts. First, a data flow analysis shows where each piece of identity data resides – Ethereum contract storage, Holochain DHT, or local wallet – at each step of the protocol, using mermaid sequence diagrams to trace registration, approval, transfer, and regulatory compliance flows. Second, a complete worked cryptographic example implements every step in Python on the `alt_bn128` (BN254) curve – executing on the compiled `buck-identity` production kernel (`buck_core.buck_identity`, arkworks-backed) through the `alberta_buck.wallet` primitives, with the pure-Python `py_ecc` path retained as the executable specification – with counter-examples demonstrating what each verification check looks like when it fails.

The scenario is a transfer from Private Alice to Public Bob (an AMM pool). Alice obtains a Pointcheval-Sanders signed identity from an issuer, derives a hiding presentation of it via the Identity Fountain, registers it on-chain (443,892 gas measured for the NIZK verification), and re-encrypts her identity for Bob with a Chaum-Pedersen proof of correct re-encryption. Bob verifies Alice’s identity using the two-channel design: the on-chain ElGamal ciphertext provides cryptographic binding, while Holochain delivers the plaintext `identity_data`. The AMM pool scenario demonstrates how this proof chain protects Bob under subpoena – even 18 months and 500,000 transactions later – and why Alice cannot hide from regulatory recovery despite the presentation’s hiding of her identity from everyone but her counterparties. The presentation is the A’ design implemented on branch `feature/a-prime`; its security argument awaits independent review. (PDF, Text)

# Contents

|           |                                                                          |           |
|-----------|--------------------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Data Flow: Where Everything Lives</b>                                 | <b>4</b>  |
| 1.1       | Data Residency . . . . .                                                 | 4         |
| 1.2       | Registration: Credential Goes On-Chain . . . . .                         | 4         |
| 1.3       | Approve: Re-Encryption with Trust Verification . . . . .                 | 5         |
| 1.4       | Transfer: Everything On-Chain . . . . .                                  | 6         |
| 1.5       | Two-Channel Verification and Malfeasance Detection . . . . .             | 7         |
| 1.6       | The Four Transfer Modes . . . . .                                        | 8         |
| 1.7       | The AMM Pool Scenario: Retroactive Identity Recovery . . . . .           | 9         |
| <b>2</b>  | <b>Setup: The <math>\text{alt\_bn128}</math> Curve and Helpers</b>       | <b>13</b> |
| <b>3</b>  | <b>Issuer Key Generation</b>                                             | <b>14</b> |
| 3.1       | Same Operation via the Wallet API . . . . .                              | 14        |
| <b>4</b>  | <b>KYC Ceremony – Issuer Signs Alice’s Identity</b>                      | <b>15</b> |
| 4.1       | Alice’s Plaintext Identity . . . . .                                     | 15        |
| 4.2       | Compute Identity Scalar and PS Signature . . . . .                       | 15        |
| 4.3       | Verify the PS Signature (Issuer Self-Check) . . . . .                    | 16        |
| 4.4       | Same Operation via the Wallet API . . . . .                              | 16        |
| <b>5</b>  | <b>The Identity Fountain – Alice Derives a Fresh Account</b>             | <b>17</b> |
| 5.1       | Present the Credential: $(A, B)$ Hides $m$ . . . . .                     | 17        |
| 5.2       | Generate Fresh Identity Key Pair and ElGamal Encryption . . . . .        | 18        |
| 5.3       | NIZK Proof: Binding the Presentation to the ElGamal Ciphertext . . . . . | 19        |
| <b>6</b>  | <b>Registration – On-Chain Verification</b>                              | <b>19</b> |
| 6.1       | What the Chain Sees, and What It Can Learn . . . . .                     | 19        |
| 6.2       | Verify the NIZK Proof (What the Contract Computes) . . . . .             | 20        |
| <b>7</b>  | <b>Bob Binds His Pool as a Public-Identity Contract</b>                  | <b>22</b> |
| 7.1       | Same Operation via the Wallet API . . . . .                              | 23        |
| <b>8</b>  | <b>Alice Approves Bob – Re-Encryption + Chaum-Pedersen Proof</b>         | <b>23</b> |
| 8.1       | Re-Encrypt Alice’s Identity for Bob . . . . .                            | 23        |
| 8.2       | Chaum-Pedersen Proof of Correct Re-Encryption . . . . .                  | 24        |
| 8.3       | Verify Chaum-Pedersen Proof (What the Contract Computes) . . . . .       | 24        |
| <b>9</b>  | <b>Transfer – Bilateral Identity Check</b>                               | <b>26</b> |
| <b>10</b> | <b>Bob Verifies Alice’s Identity (Two-Channel Design)</b>                | <b>27</b> |
| 10.1      | Counter-example: Tampered Identity Data . . . . .                        | 27        |
| <b>11</b> | <b>Receipt Construction</b>                                              | <b>28</b> |
| <b>12</b> | <b>Summary: What Each Party Knows</b>                                    | <b>28</b> |
| <b>13</b> | <b>Appendix: The Identity Fountain – The Two-Sided Linkability Test</b>  | <b>29</b> |

|                                                                  |           |
|------------------------------------------------------------------|-----------|
| <b>14 Appendix: Small-Group Proofs</b>                           | <b>30</b> |
| 14.1 Setup and ElGamal . . . . .                                 | 31        |
| 14.2 NIZK: Proof of Knowledge (Registration) . . . . .           | 32        |
| 14.3 Chaum-Pedersen: Proof of Same Plaintext (Approve) . . . . . | 33        |

# 1 Data Flow: Where Everything Lives

The Alberta Buck identity system spans two networks (Ethereum and Holochain) and a local device (the wallet). The tables and mermaid sequence diagrams below map every piece of identity data to its storage location, visibility, and the protocol step that consumes it – tracing flows through registration, approval, transfer, and two-channel verification, including the trust checkpoint where wallets detect malfeasance, and the AMM pool scenario where a public pool operator responds to a regulatory subpoena months after a swap. The key result: all cryptographic proofs are permanently on-chain, so regulatory compliance requires only Bob’s secret key and an archival Ethereum node – not Alice’s cooperation.

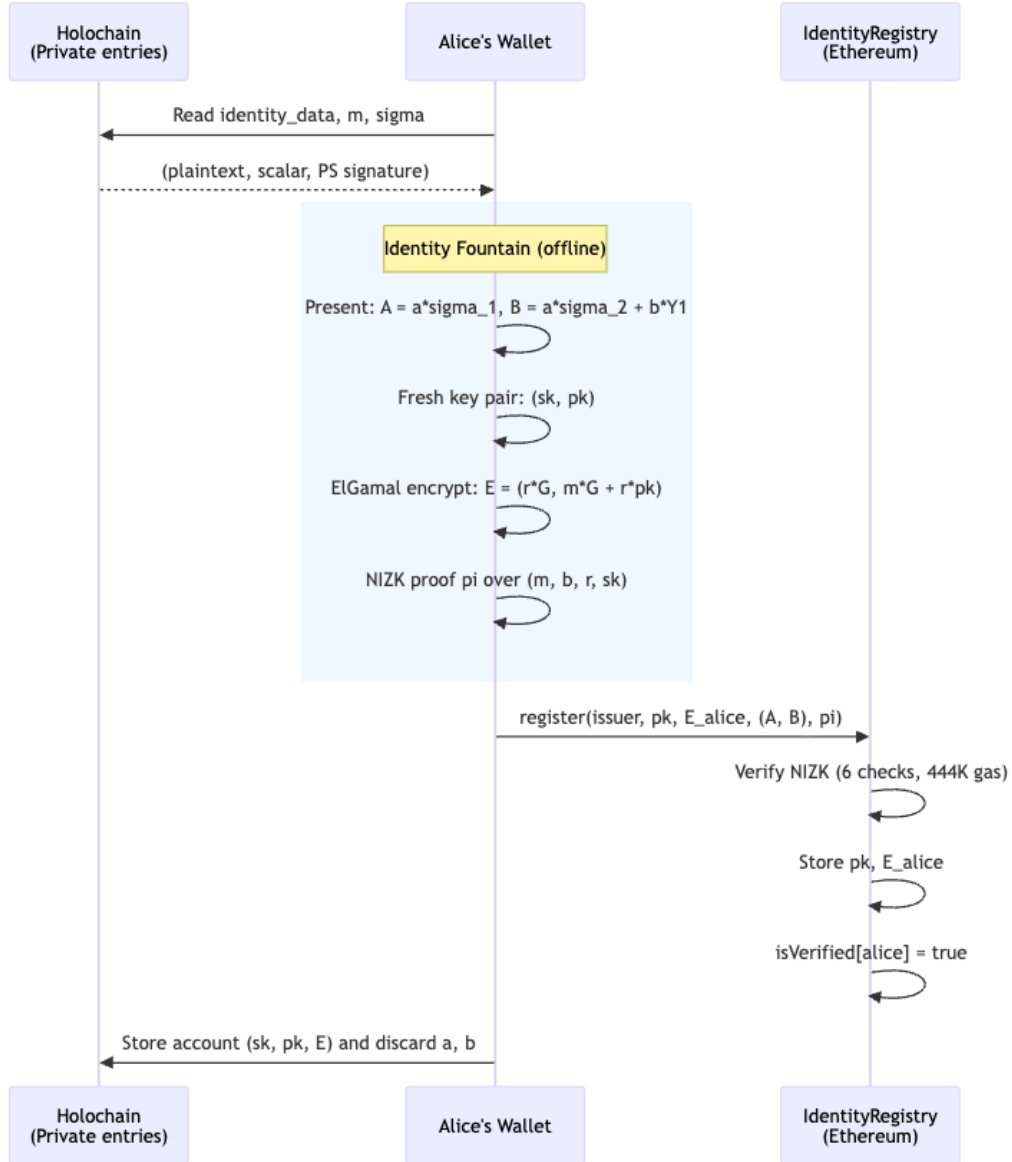
## 1.1 Data Residency

| Data                           | Location                 | Visibility        | When Used                            |
|--------------------------------|--------------------------|-------------------|--------------------------------------|
| identity_data = plaintext      | Holochain Private entry  | Owner only        | Approve (sent to counterparty)       |
| $m = H(\text{identity\_data})$ | Holochain Private entry  | Owner only        | Registration, Approve                |
| PS signature $\sigma$          | Holochain Private entry  | Owner only        | Registration (blinded into (A, B))   |
| Blinding $b$ (per account)     | Wallet memory            | Owner only        | Registration proof, then discarded   |
| Secret key $sk$                | Holochain Private entry  | Owner only        | Approve (decrypt + re-encrypt)       |
| Presentation (A, B)            | IdentityRegistry (tx)    | Public (on-chain) | Registration verification            |
| Public key $pk$                | IdentityRegistry         | Public (on-chain) | Approve, Transfer                    |
| ElGamal ciphertext $E_{alice}$ | IdentityRegistry         | Public (on-chain) | Approve (read by contract)           |
| NIZK proof $\pi$               | IdentityRegistry (tx)    | Public (on-chain) | Registration verification            |
| Re-encrypted $E_{bob}$         | BUCK ERC-20 (approve)    | Public (on-chain) | Two-channel verification             |
| Chaum-Pedersen proof           | BUCK ERC-20 (approve tx) | Public (on-chain) | Approve verification                 |
| _receiptFragments[a] [b]       | BUCK ERC-20 storage      | Public (on-chain) | Transfer (bilateral check)           |
| isPublicIdentity flag          | IdentityRegistry         | Public (on-chain) | Transfer (receipt-fragment fallback) |
| Public identity_data           | IdentityRegistry or DHT  | Public            | Anyone (no approve needed)           |
| Holochain Warrants             | Holochain DHT            | Public            | Malfeasance detection                |
| Canonical receipt              | Holochain Private entry  | Each party only   | Dispute resolution                   |

This table says where each item lives. What the public items can *reveal* – why a permanently archived registration transaction discloses no identity to any node – is set out item by item in What the Chain Sees, and What It Can Learn.

## 1.2 Registration: Credential Goes On-Chain

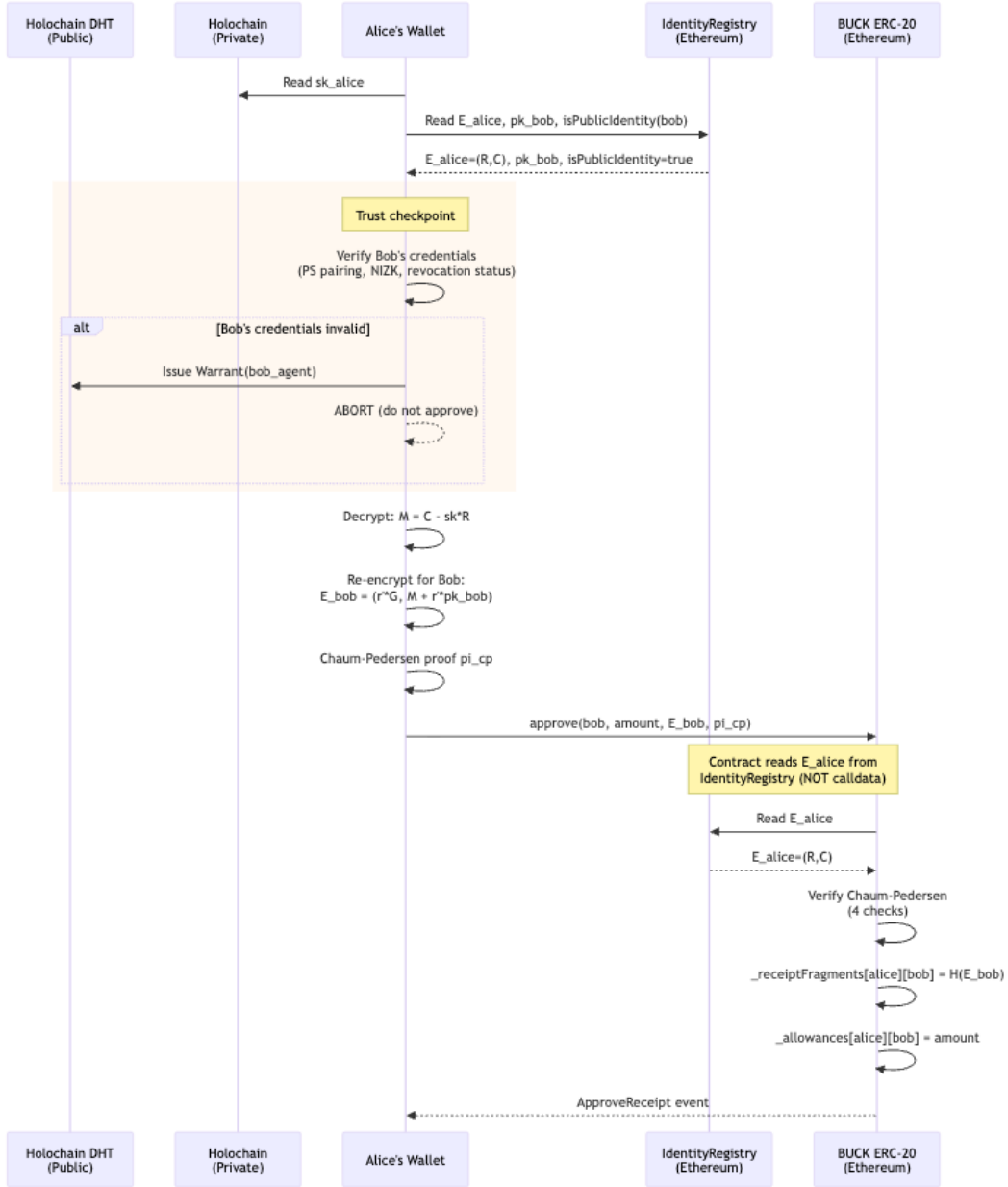
Alice’s wallet reads her Private entries, derives a *presentation* of her credential entirely offline, then submits it to the IdentityRegistry. The contract verifies the NIZK proof (443,892 gas measured on the A’ branch with an in-process EVM) and stores the public artifacts. The secret key, the raw PS signature and the plaintext identity *never* touch Ethereum; the presentation (A, B) that does is a uniform pair of points that no holder of the identity – the issuer included – can test.



### 1.3 Approve: Re-Encryption with Trust Verification

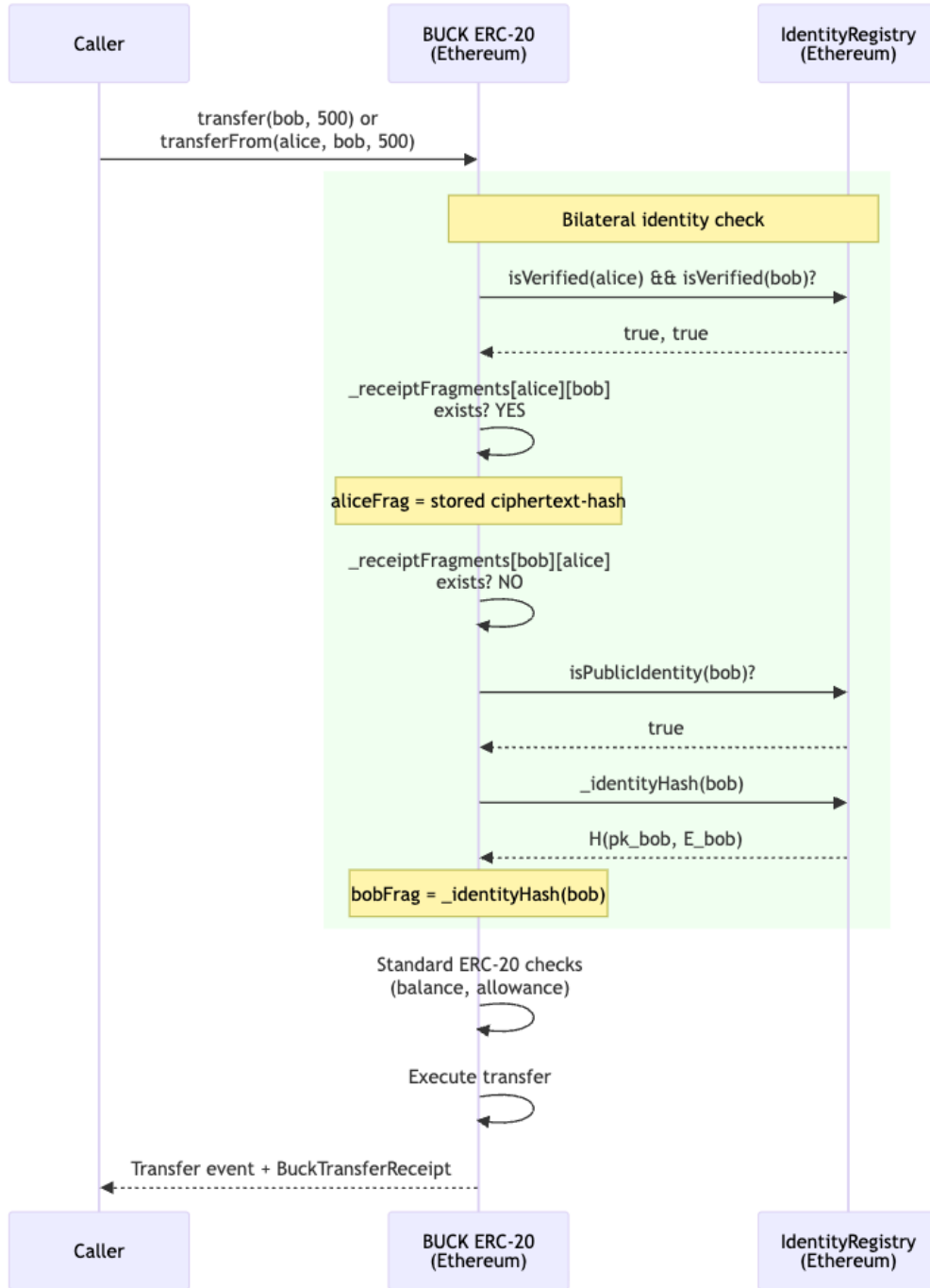
When Alice wants to transact with Bob, her wallet first *verifies Bob's credentials* before re-encrypting. This is the critical trust checkpoint: Alice's wallet reads Bob's on-chain data, checks his PS signature (via the pairing equation) and public key, and only then produces the re-encrypted ciphertext.

If anything is wrong – invalid PS signature, mismatched NIZK proof, revoked credential – Alice simply *does not call approve*. No transaction, no risk. If Alice's wallet detects provable malfeasance (e.g., a forged PS signature from a revoked issuer), it can issue a Holochain **Warrant** against Bob's agent. Multiple independent wallets detecting the same fraud produce consensus: the warranted agent becomes untrusted on the DHT.



## 1.4 Transfer: Everything On-Chain

By the time `transfer` or `transferFrom` is called, all proofs have already been verified. The EVM code only checks boolean flags and stored receipt fragments – no cryptography runs during the transfer itself. Public-Identity contracts (those bound via `bindContract(target, pk, E, isPublicIdentity_=true)`) need not have a per-pair receipt fragment: the receipt falls back to the bound `_identityHash` for that side.

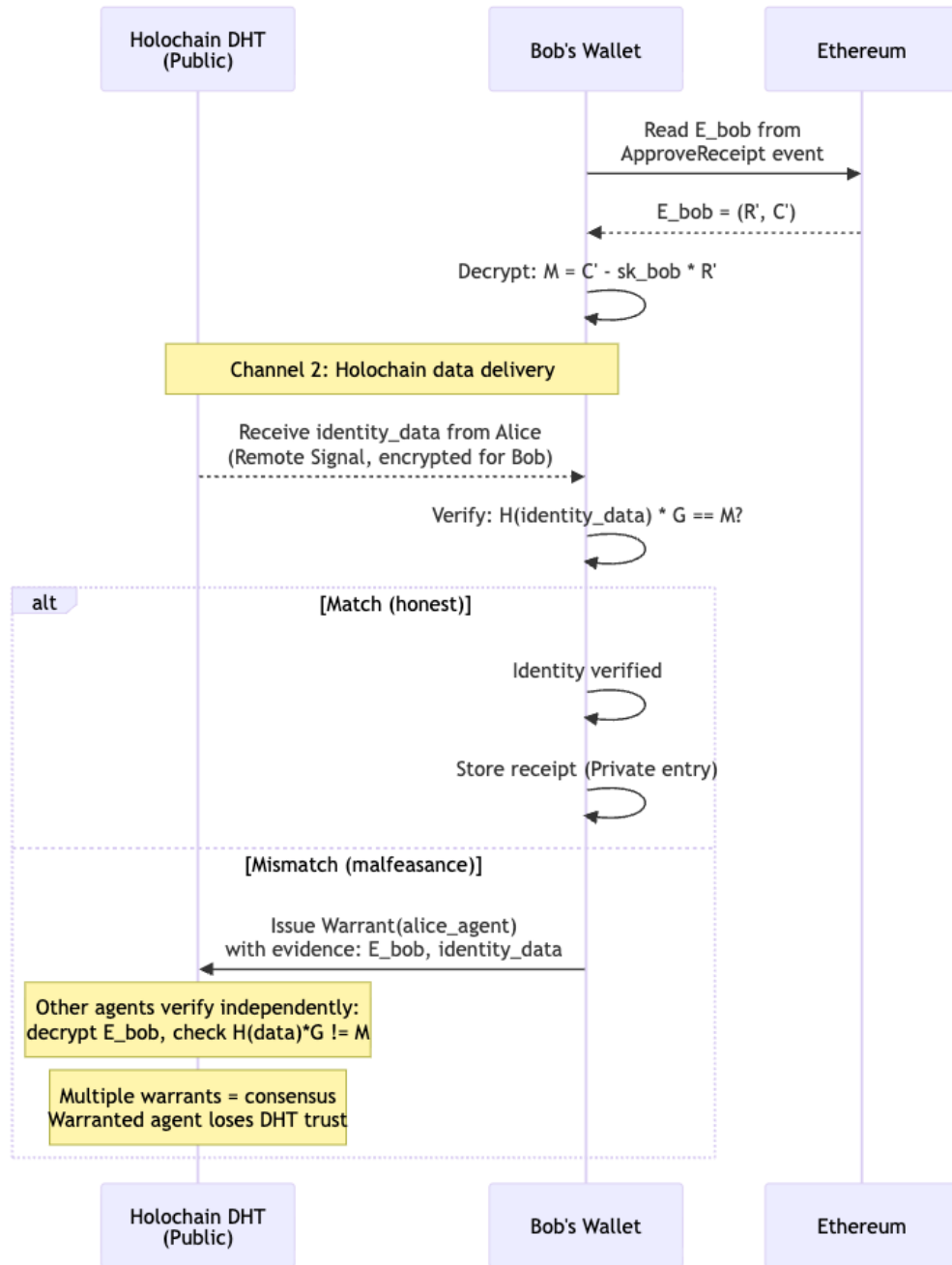


## 1.5 Two-Channel Verification and Malfeasance Detection

After the transfer, Bob verifies Alice's identity using two independent channels. Channel 1 (Ethereum) provides the binding proof; Channel 2 (Holochain) provides the content. Neither channel alone is sufficient – this is what makes the design tamper-evident.

If Bob detects that the off-chain `identity_data` doesn't match the on-chain point  $M$ , this is *provable* malfeasance: Bob holds the ciphertext (on-chain, timestamped) and the mismatched data (signed by Alice's Holochain agent). Bob issues a Warrant; other agents independently verify the

evidence. Since Holochain agents sign their source chain entries, Alice cannot deny having sent the bad data.



## 1.6 The Four Transfer Modes

The amount of pre-work depends on the visibility of each party:

| Sender  | Receiver | Who Calls <code>approve</code> | Holochain Work        | On-Chain Gas        |
|---------|----------|--------------------------------|-----------------------|---------------------|
| Private | Private  | Both (Alice→Bob, Bob→Alice)    | Both exchange via DHT | 2 x ~29K            |
| Private | Public   | Alice only                     | Alice sends via DHT   | 1 x ~29K            |
| Public  | Private  | Bob only                       | Bob sends via DHT     | 1 x ~29K            |
| Public  | Public   | Neither                        | None                  | 0 (standard ERC-20) |

In every case, the `transfer` / `transferFrom` call itself does *no* cryptography – it only reads the boolean results of prior `approve` calls and the `isPublicIdentity` flag bound at `bindContract` time. All proof verification happens during `approve`.

## 1.7 The AMM Pool Scenario: Retroactive Identity Recovery

Consider Bob’s AMM liquidity pool – a public BUCK account that processes swaps automatically. Over its lifetime, a million private accounts call `approve(pool, amount, E_pool, proof)` and then swap via `transferFrom`. Bob’s pool contract never examines any counterparty’s identity at the time of the swap. It doesn’t need to: the EVM already verified every proof.

Eighteen months later, a regulator subpoenas Bob for Alice’s identity – account #500,000 – in connection with a fraud investigation. How does Bob comply, and how does the on-chain record prove his pool wasn’t complicit?

### 1.7.1 What the Chain Already Guarantees

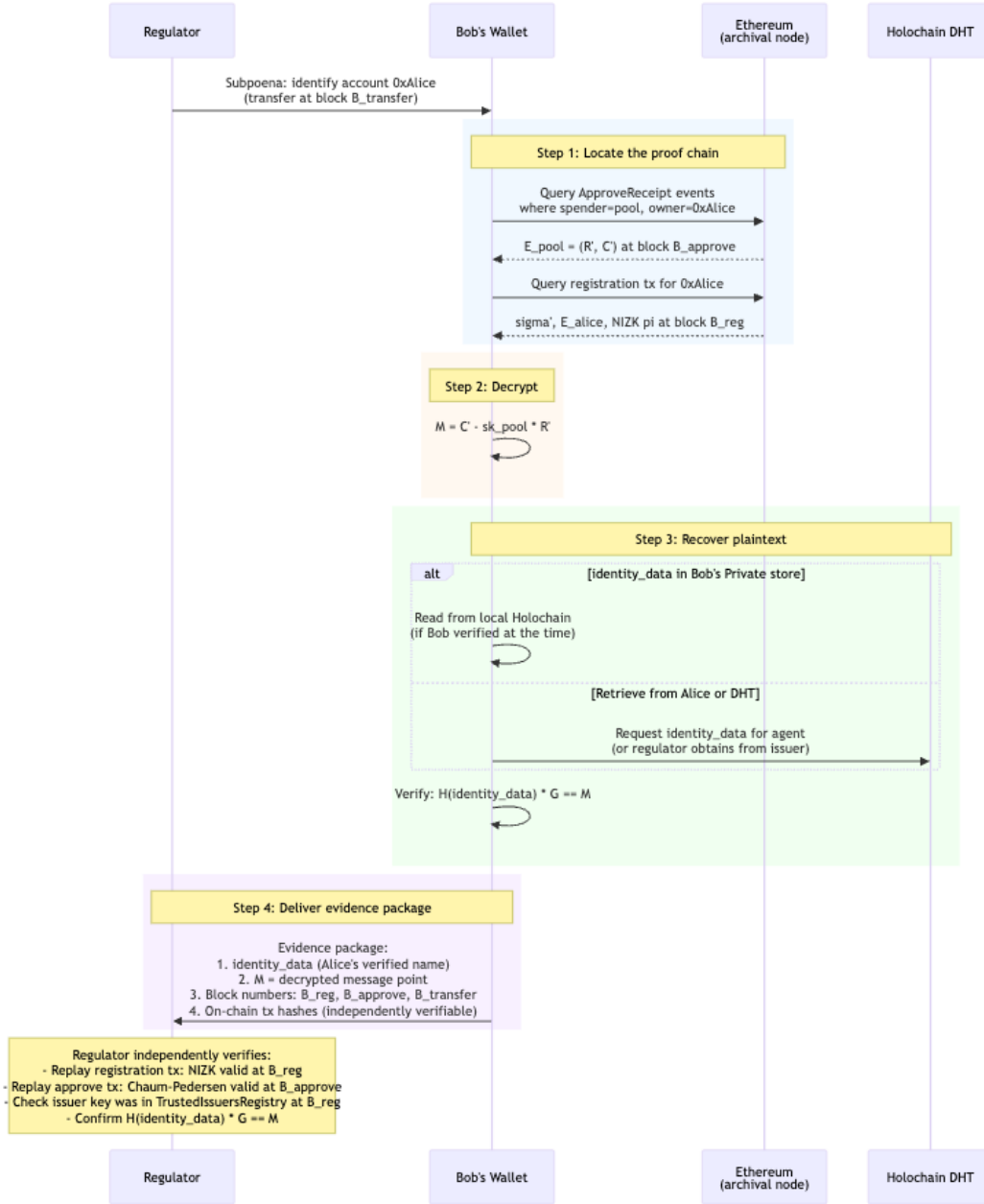
Every `approve` that succeeded is an *immutable proof receipt*. The EVM verified the Chaum-Pedersen proof at that block height; the IdentityRegistry verified Alice’s NIZK at registration time. These facts are encoded in the transaction receipts themselves – the transactions *could not have succeeded* without valid proofs.

The chain of cryptographic commitments is:

1. **Issuance:** A trusted issuer (trusted by the IdentityRegistry) signed Alice’s identity scalar  $m$  with a PS signature  $\sigma$ . The issuer’s public key was on-chain and valid at that epoch.
2. **Registration:** The IdentityRegistry verified Alice’s NIZK proof – binding  $\sigma'$  (rerandomized PS signature) to  $E_{\text{alice}}$  (ElGamal ciphertext). The NIZK proves the *same*  $m$  appears in both. The transaction succeeded at block  $B_{\text{reg}}$ ; the NIZK was valid at that block.
3. **Approve:** The BUCK contract verified Alice’s Chaum-Pedersen proof – proving  $E_{\text{pool}}$  encrypts the *same* message point  $M$  as her registered  $E_{\text{alice}}$ . The contract read  $E_{\text{alice}}$  from the IdentityRegistry (not from calldata), so Alice couldn’t substitute a different ciphertext. The transaction succeeded at block  $B_{\text{approve}}$ .
4. **Transfer:** The bilateral identity check passed. The pool was bound under a Public Identity (`isPublicIdentity(pool) = true` via `IdentityRegistry.bindContract`), and Alice had a valid `_receiptFragments[alice][pool]` from her `approve`. The pool side fell back to the bound `_identityHash(pool)`. The swap executed at block  $B_{\text{transfer}}$ .

None of these steps required Bob to be online, to run a Holochain node, or to examine Alice’s identity. The EVM did it for him.

### 1.7.2 Bob Responds to the Subpoena



### 1.7.3 Why Bob Is Protected

Bob's defense is the proof chain itself. He doesn't need to argue that he checked Alice's identity – the *contract* checked it, and the Ethereum transaction log is the evidence:

- **Registration tx succeeded**  $\Rightarrow$  NIZK was valid  $\Rightarrow$  a trusted issuer certified  $m$  at that epoch.
- **Approve tx succeeded**  $\Rightarrow$  Chaum-Pedersen was valid  $\Rightarrow E_{\text{pool}}$  encrypts the same  $M$  as  $E_{\text{alice}}$ .
- **Bob decrypts**  $E_{\text{pool}}$  to  $M$ ;  $M = H(\text{identity\_data}) \cdot G$ .

The regulator can independently replay these transactions against an archival node. No trust in Bob is required – the math is self-certifying.

Even in adversarial scenarios, Bob is covered:

| Scenario                             | Why Bob is protected                                                                                                                                      |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alice used a stolen identity         | The <i>issuer</i> certified it; Bob’s contract verified the issuer’s PS signature was valid at that epoch. Liability falls on the issuer’s KYC process.   |
| Alice’s issuer key was later revoked | The registration tx is timestamped. The issuer key was trusted by the IdentityRegistry at block $B_{\text{reg}}$ .                                        |
| Alice’s identity epoch expired       | Epoch expiry is enforced at registration time. If the epoch was valid then, the proof stands.                                                             |
| Alice’s Holochain agent is offline   | Bob needs only $E_{\text{pool}}$ (on-chain) and $\text{sk\_pool}$ (his own key). The plaintext can be obtained from the issuer if Alice is uncooperative. |
| Alice denies the transaction         | The approve tx is signed by Alice’s Ethereum account. Non-repudiation is inherent in the blockchain.                                                      |

### 1.7.4 Data Availability Over Time

The critical question: will Bob still have what he needs 18 months later?

| Data                                 | Availability                                                                | Notes                                      |
|--------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------|
| $E_{\text{pool}}$ (R’,C’)            | Permanent (Ethereum event log)                                              | ApproveReceipt event                       |
| $E_{\text{alice}}$                   | Permanent (IdentityRegistry storage)                                        | Until account is closed                    |
| $\sigma'$ , NIZK $\pi$               | Permanent (registration tx calldata)                                        | Archival node required                     |
| Chaum-Pedersen proof                 | Permanent (approve tx calldata)                                             | Archival node required                     |
| $\text{sk\_pool}$ (Bob’s secret key) | Bob’s custody                                                               | Standard key management                    |
| $\text{identity\_data}$ (plaintext)  | Holochain DHT (if agent is online)<br>or Bob’s Private store (if he cached) | May require issuer cooperation as fallback |
| Issuer public key history            | IdentityRegistry trusted-issuer table                                       | Includes revocation timestamps             |

Everything except the plaintext  $\text{identity\_data}$  is permanently on-chain. The plaintext is the one piece that lives off-chain – but Bob only needs it to produce a *human-readable name* for the regulator. The cryptographic proof chain stands without it:  $M$  is recoverable from  $E_{\text{pool}}$  alone, and  $M$  is what the issuer certified.

If Alice’s Holochain agent is long gone, the regulator can obtain  $\text{identity\_data}$  from the issuer (who has it from the original KYC ceremony) and verify  $H(\text{identity\_data}) \cdot G = M$  independently. The issuer doesn’t need Bob’s cooperation; the regulator doesn’t need Alice’s.

### 1.7.5 Why $M$ Is the Same Across All Accounts – And Why That’s Correct

A natural concern: if Alice derives ten accounts from the Identity Fountain, and each counterparty who decrypts their re-encrypted ciphertext recovers the same  $M = m \cdot G$ , doesn’t that break unlinkability?

No. The Identity Fountain’s guarantee is about *on-chain unlinkability* – what a passive observer (Eve) can learn from the public blockchain data alone, without any secret keys. That guarantee is airtight:

- $\sigma'_1$  and  $\sigma'_2$  are *statistically* unlinkable (information-theoretic, not merely computational)
- $E_1$  and  $E_2$  are IND-CPA secure under independent keys
- $pk_1$  and  $pk_2$  are independent random points

No algorithm, no amount of computation, can correlate Alice’s accounts from on-chain data.

The identity scalar  $m = H(\text{identity\_data})$  is *intentionally invariant* because it *is* Alice’s identity. If each account used a different  $m$ , the two-channel verification would break: Bob couldn’t verify that the **identity\_data** he received matches what the issuer certified. The Identity Fountain protects the *credential wrapping* ( $\sigma'$ ,  $pk$ ,  $E$ ), not the identity itself.

Think of it as giving Alice a new disguise for each account – different keys, different signature, different ciphertext. The disguises fool passive surveillance (Eve). They don’t fool someone Alice voluntarily shows her face to (Bob, during **approve**).

| Observer                | Can link accounts? | Why?                                               |
|-------------------------|--------------------|----------------------------------------------------|
| Eve (passive, on-chain) | NO                 | All public artifacts are independent random values |
| Bob (authorized)        | Learns one account | By design – the point of <b>approve</b>            |
| Bob + Carol (colluding) | Yes, via $M$       | But both already know Alice’s name                 |
| Issuer (on-chain only)  | NO                 | Can’t extract $M$ without a secret key             |
| Issuer (given $M$ )     | YES                | By design – regulatory compliance path             |

The privacy boundary is clear: *before approve*, Alice’s accounts are unlinkable to everyone. *After approve*, the specific counterparty learns Alice’s identity for that account – and only that counterparty (plus anyone with whom they share  $M$ , but they already shared Alice’s name via the **identity\_data** exchange).

#### 1. Alice Tries to Hide by Not Publishing **identity\_data**

Alice might try to game the system: complete the on-chain **approve** (re-encryption + Chaum-Pedersen proof), satisfying the ERC-20’s bilateral identity check, but never publish her **identity\_data** to the Holochain DHT. She hopes that if a subpoena arrives, Bob won’t be able to produce a human-readable name.

This doesn’t work. Bob still holds the cryptographic thread:

- Bob decrypts  $E_{\text{pool}}$  to recover  $M = m \cdot G$ . This requires only Bob’s secret key and the on-chain ciphertext – Alice’s cooperation is unnecessary.
- The regulator takes  $M$  to the issuer. The issuer computes  $m_i \cdot G$  for each identity they have issued and finds the match. The issuer has Alice’s **identity\_data** from the original KYC ceremony.
- The issuer confirms: this  $M$  belongs to Alice Johnson.

Alice’s attempt to hide by not publishing is futile: the on-chain  $E_{\text{pool}}$  and the issuer’s records are sufficient. Alice cannot retroactively alter either.

#### 2. Bob’s Wallet Detects Non-Publication

Bob’s wallet (Holochain hApp) can proactively monitor for counterparties who complete the on-chain **approve** but fail to publish the required DHT entries. The expected protocol is:

- (a) Alice calls `approve(pool, amount, E_pool, proof)` on Ethereum.
- (b) Alice’s wallet publishes a DHT entry containing her `identity_data`, encrypted for Bob’s Holochain agent, and keyed by the `ApproveReceipt` event hash.
- (c) Bob’s wallet watches for the `ApproveReceipt` event and looks up the corresponding DHT entry within a grace period.

If step 2 never happens – the on-chain `approve` succeeded but no DHT entry appears – Bob’s wallet can:

- Flag the account as non-compliant (Alice approved on-chain but didn’t complete the Holochain identity exchange).
- Issue a Holochain **Warrant** against Alice’s agent, citing the on-chain tx hash as evidence of protocol violation.
- Report the account to pool operators or compliance systems.
- Optionally, Bob’s contract could revoke the approval (`reset_receiptFragments[alice][pool]`), blocking future swaps until Alice completes the full protocol.

The warrant is independently verifiable: any Holochain agent can check that the `ApproveReceipt` event exists on Ethereum but the corresponding DHT entry does not. Multiple independent warrants produce DHT-level consensus that Alice’s agent is non-compliant.

Even if Alice never publishes, Bob’s regulatory position is unchanged. He can still decrypt *M* from the on-chain ciphertext and hand it to the regulator, who obtains `identity_data` from the issuer. Alice’s non-publication delays the human-readable identification by one step (issuer lookup) but cannot prevent it.

## 2 Setup: The `alt_bn128` Curve and Helpers

All operations use BN254, the curve behind Ethereum’s `ecAdd` (0x06), `ecMul` (0x07), and `ecPairing` (0x08) precompiles. The `alberta_buck.wallet.bn254` primitives provide exactly these operations, executing on the compiled `buck-identity` kernel (`buck_core.buck_identity`, arkworks BN254) – the first block asserts it, so this worked example demonstrates the production cryptography. The pure-Python `py_ecc` path remains the executable specification the kernel is proven bit-identical against (`core/vectors/identity-kernel-vectors.json`). Helper functions for seeded random scalar generation (one deterministic transcript, so the printed `RESULTS` reproduce byte for byte), `keccak256` hashing, and point formatting are defined here; all subsequent code blocks share a single `:session` `abid` so that variables persist across steps.

```
import random
from web3 import Web3
from alberta_buck.wallet import backend
from alberta_buck.wallet.bn254 import (
    G1,          # Generator of G_1
    G2,          # Generator of G_2
    Z1,          # Point at infinity (G_1 identity)
    ORDER,       # Curve group order (prime)
    mul,         # Scalar multiplication in G_1 or G_2
    add,         # Point addition
    neg,         # Point negation
    eq,          # Point equality
    pairing,     # Optimal Ate pairing: G_2 x G_1 -> G_T
```

```

    is_inf, # Point-at-infinity test
    FQ12_one, # G_T identity (pairing-product checks)
)

assert backend() == "kernel", "worked example must run the production kernel"

_seed = random.Random(0xAB1DE7A1) # ONE deterministic transcript

def rand():
    """Random non-zero scalar mod ORDER (seeded; the transcript reproduces)."""
    while True:
        v = _seed.getrandbits(256) % ORDER
        if v:
            return v

def keccak(*args):
    """keccak256 over concatenated big-endian 32-byte words, reduced mod ORDER."""
    data = b''.join(v.to_bytes(32, 'big') if isinstance(v, int)
                    else str(v).encode() for v in args)
    h = Web3.solidity_keccak(['bytes'], [data])
    return int.from_bytes(h, 'big') % ORDER

def pt(P, label=""):
    """Format a G_1 point for display."""
    if is_inf(P):
        return f"{label}0 (infinity)"
    return f"{label}({int(P[0]) % 10**6:>6d}..., {int(P[1]) % 10**6:>6d}...)"

print(f"wallet crypto backend: {backend()} (buck_core.buck_identity)")
print(f"BN254 curve order: {ORDER}")
print(f"G_1 generator:      {pt(G1)}")
print(f"G_2 generator:      (pair of FQ2 elements)")

```

### 3 Issuer Key Generation

The identity issuer (e.g., ATB Financial acting as Alberta's KYC authority) generates a Pointcheval-Sanders key pair: secret key  $(x, y)$  (two scalars), public key  $(X, Y, Y_1)$  (two points in  $G_2$  plus the  $G_1$  image  $Y_1 = y \cdot G$  that holders blind with, stored on-chain by `IdentityRegistry.trustIssuer`, which checks  $e(Y_1, g_2) = e(G, Y)$  once). This key pair is the root of trust for all identities the issuer certifies; every credential in the system traces back to a PS signature under this key.

```

# Issuer's PS secret key
isk_x = rand()
isk_y = rand()

# Issuer's PS public key: (X, Y) in G_2 plus Y1 = y*G in G_1, the base
# holders blind with. IdentityRegistry.trustIssuer stores all three and
# checks e(Y1, g_2) == e(G, Y) once.
IPK_X = mul(G2, isk_x)
IPK_Y = mul(G2, isk_y)
IPK_Y1 = mul(G1, isk_y)
assert pairing(G2, IPK_Y1) == pairing(IPK_Y, G1)

print("Issuer PS key pair generated.")
print(f"  sk = (x, y)          -- two secret scalars")
print(f"  PK = (X, Y, Y1)        -- X, Y in G_2; Y1 = y*G in G_1 (trustIssuer checks it against Y)")

```

#### 3.1 Same Operation via the Wallet API

The reference implementation in `alberta_buck.wallet.issuer` packages the same key generation behind a typed `Issuer` class. Wrapping the scalars produced above into an `Issuer` lets every later step in this document call the wallet API while keeping the keys identical to the raw-crypto example.

```
from alberta_buck.wallet import Issuer, PSKeyPair

ATB_ADDR = 0xa7b00000000000000000000000000000a7b00
issuer_obj = Issuer(
    issuer_id="atb-financial-ca",
    issuer_addr=ATB_ADDR,
    keypair=PSKeyPair(sk_x=isk_x, sk_y=isk_y, pk_X=IPK_X, pk_Y=IPK_Y, pk_Y1=IPK_Y1),
)

from alberta_buck.wallet import ps_key_consistent
assert ps_key_consistent(issuer_obj.pk_X, issuer_obj.pk_Y, issuer_obj.pk_Y1)
print(f"Wallet API: {issuer_obj.issuer_id!r} @ 0x{issuer_obj.issuer_addr:040x}")
print(f"    pk_X, pk_Y, pk_Y1 match the raw points above; ps_key_consistent: True")
```

#### 4 KYC Ceremony – Issuer Signs Alice’s Identity

Alice presents her identity documents to the issuer, who verifies them, constructs a canonical `identity_data` JSON structure (sorted keys, compact separators, raw UTF-8 – the one canonical JSON dialect, shared with the receipt envelope), and computes the identity scalar  $m = \text{keccak256}(\text{identity\_data}) \bmod \text{ORDER}$ . The issuer then produces a PS signature  $\sigma = (h, (x + m \cdot y) \cdot h)$  binding  $m$  to the issuer’s secret key. The bilinear pairing equation verifies the signature, and a counter-example confirms that any change to `identity_data` (producing a different  $m$ ) causes verification to fail.

### 4.1 Alice's Plaintext Identity

```
import json

# Alice's verified identity (canonical encoding -- field order matters)
alice_identity_data = json.dumps({
    "given_name": "Alice",
    "family_name": "Johnson",
    "jurisdiction": "Alberta, Canada",
    "id_type": "Alberta Identity Card",
    "id_number": "AIC-2026-4839201",
    "date_of_birth": "1992-03-15",
    "issuer_id": "atb-financial-ca",
    "issued_at": "2026-01-20T14:30:00Z",
    "epoch": 42
}, sort_keys=True, separators=(',', ':'))

print("Alice's identity_data (canonical JSON):")
print(f" {alice_identity_data[:72]}...")

Alice's identity_data (canonical JSON):
{"date_of_birth": "1992-03-15", "epoch": 42, "family_name": "Johnson", "given_...
```

## 4.2 Compute Identity Scalar and PS Signature

The PS signature is  $\sigma = (h, (x + m \cdot y) \cdot h)$  where  $h$  is a random point in  $G_1$ .

```
# Identity scalar:  $m = \text{keccak256}(\text{identity\_data}) \bmod \text{ORDER}$ 
m_alice = keccak(alice_identity_data)

# PS signature: pick random h, compute  $\sigma = (h, (x + m \cdot y) \cdot h)$ 
t_h      = rand()
h         = mul(G1, t_h)                # random G_1 point
sigma_2   = mul(h, (isk_x + m_alice * isk_y) % ORDER)
sigma     = (h, sigma_2)

print(f"Identity scalar m = H(identity_data):")
```

```

print(f" m = {m_alice}")
print(f"PS signature sigma:")
print(f" sigma_1 = {pt(sigma[0])}")
print(f" sigma_2 = {pt(sigma[1])}")

```

### 4.3 Verify the PS Signature (Issuer Self-Check)

The PS verification equation uses a bilinear pairing:

$$e(\sigma_1, X + m \cdot Y) = e(\sigma_2, g_2)$$

**What this proves:** The issuer's signature is mathematically valid – it binds the identity scalar  $m$  to the issuer's secret key. If the signature passes, Alice can be sure the issuer actually certified her identity (no one else knows  $(x, y)$ ).

```

# Verify: e(sigma_1, X + m*Y) == e(sigma_2, g_2)
lhs = pairing(add(IPK_X, mul(IPK_Y, m_alice)), sigma[0])
rhs = pairing(G2, sigma[1])
assert lhs == rhs, "PS signature verification FAILED"
print("PS signature verified: e(sigma_1, X + m*Y) == e(sigma_2, g_2)")

# The issuer gives Alice: (m, sigma, identity_data)
# Alice stores all three in her Holochain Private entry
print(f"\nIssuer gives Alice:")
print(f" m = {m_alice}")
print(f" sigma = (sigma_1, sigma_2) -- PS signature")
print(f" identity_data = '{alice_identity_data[:40]}...'")

```

**How to read the output:** "PS signature verified" means both sides of the pairing equation produce the same element in  $G_T$ . If  $m$  were wrong (e.g., tampered identity data), the two pairings would produce *different*  $G_T$  elements and the assertion would fail. A counter-example follows.

#### 4.3.1 Counter-example: Wrong Identity Scalar

If Eve tries to forge a signature using a different  $m$ , the pairing equation fails. This is what a rejected verification looks like:

```

# Tampered m -- what if someone changes one field in identity_data?
m_fake = keccak("tampered_identity_data")
lhs_bad = pairing(add(IPK_X, mul(IPK_Y, m_fake)), sigma[0])
rhs_good = pairing(G2, sigma[1])
ok = (lhs_bad == rhs_good)
print(f"PS verify with wrong m: {ok}")
print(f" (The two G_T elements differ -- the pairing detects the mismatch.)")
print(f" Any change to identity_data produces a different m, which")
print(f" makes the pairing equation fail. The signature is unforgeable.")

```

### 4.4 Same Operation via the Wallet API

`Issuer.issue(...)` bundles the canonicalization,  $m$  computation, and PS signing into one call. `ps_verify(...)` performs the same pairing equation as the raw block above; the counter-example uses the same primitive against a tampered  $m$ .

```

from alberta_buck.wallet import ps_verify, identity_scalar
ALICE_ADDR = 0xa11ce00000000000000000000000000000a11ce

alice_fields = {
    "given_name": "Alice",
    "family_name": "Johnson",

```

```

    "jurisdiction": "Alberta, Canada",
    "id_type": "Alberta Identity Card",
    "id_number": "AIC-2026-4839201",
    "date_of_birth": "1992-03-15",
    "issuer_id": "atb-financial-ca",
    "issued_at": "2026-01-20T14:30:00Z",
    "epoch": 42,
}
cred = issuer_obj.issue(alice_fields, ALICE_ADDR, rng=rand) # seeded: the transcript reproduces
# Wallet computes m the same way: m = keccak256(canonical) mod ORDER
assert cred.m == identity_scalar(cred.canonical) == m_alice
# PS verification, same pairing equation as the raw block:
ok = ps_verify(issuer_obj.pk_X, issuer_obj.pk_Y, cred.sigma, cred.m)
ok_bad = ps_verify(issuer_obj.pk_X, issuer_obj.pk_Y, cred.sigma, m_fake)
print(f"Wallet API: cred.m == m_alice : {cred.m == m_alice}")
print(f" ps_verify(sigma, m_alice) : {ok}")
print(f" ps_verify(sigma, m_fake) : {ok_bad}")

```

## 5 The Identity Fountain – Alice Derives a Fresh Account

Alice wants to create a new Ethereum account. Her Holochain wallet runs the Identity Fountain entirely offline, producing four artifacts: a *presentation*  $(A, B)$  of her PS credential that hides which credential – and which identity – it presents, a fresh key pair  $(sk, pk)$ , an ElGamal encryption of  $m$  under  $pk$ , and a NIZK proof binding the presentation to the ciphertext and to the key. Each step is demonstrated below. The first block shows why the wallet publishes a presentation and not a rerandomized signature: a rerandomized signature is still a signature on  $m$ , so anyone holding  $m$  – the issuer, or any past counterparty – can test it; the presentation fails that test.

### 5.1 Present the Credential: $(A, B)$ Hides $m$

Rerandomization,  $\sigma' = (t\sigma_1, t\sigma_2)$ , yields a fresh valid signature on the same  $m$ . It is the wallet's internal re-issuance step and nothing more: the pairing test  $e(\sigma'_1, X + mY) = e(\sigma'_2, g_2)$  identifies it to anyone who knows  $m$ . What the wallet publishes is the presentation

$$A = a \cdot \sigma_1, \quad B = a \cdot \sigma_2 + b \cdot Y_1 = (x + my) \cdot A + b \cdot Y_1,$$

for fresh nonzero  $a, b$ , where  $Y_1 = y \cdot G$  is the issuer's published  $G_1$  key component.  $(A, B)$  satisfies the *masked* equation  $e(B, g_2) = e(A, X) \cdot e(mA + bG, Y)$  and no other: without  $b$  it is a uniform pair of points, the same distribution for every credential on every identity.

**What this proves:** the raw test succeeds on the rerandomized pair and fails on the presentation, for the true  $m$ ; the masked equation, which needs  $b$ , holds for the presentation.

```

from alberta_buck.wallet import ps_present

# The raw test any holder of m can run on a published pair
def raw_test(S1, S2, m):
    return pairing(add(IPK_X, mul(IPK_Y, m)), S1) == pairing(G2, S2)

# Rerandomize (wallet-internal): still a signature on m_alice
t_rerand = rand()
sigma_p = (mul(sigma[0], t_rerand), mul(sigma[1], t_rerand))
print(f"Rerandomized pair: raw PS test with m_alice -> {raw_test(sigma_p[0], sigma_p[1], m_alice)}")
print(" (a candidate test for anyone holding m: never published)")

# Present (published): (A, B) = (a*sigma_1, a*sigma_2 + b*Y1), fresh a, b
pres_alice, a_alice, b_alice = ps_present(cred.sigma, issuer_obj.pk_Y1, rng=rand)
A_alice, B_alice = pres_alice.A, pres_alice.B
print(f"Presentation A = {pt(A_alice)} B = {pt(B_alice)}")

```

```

print(f"Presentation:      raw PS test with m_alice -> {raw_test(A_alice, B_alice, m_alice)}")

# The masked equation holds -- but only with the blinding b
P_alice = add(mul(A_alice, m_alice), mul(G1, b_alice))
masked = pairing(G2, B_alice) == pairing(IPK_X, A_alice) * pairing(IPK_Y, P_alice)
print(f"Presentation:      masked equation with (m_alice, b) -> {masked}")

```

**How to read the output:** Three lines to check. The rerandomized pair passes the raw test with Alice's  $m$ : whoever holds  $m$  can identify it, which is why it stays in the wallet. The presentation fails the same test. The masked equation passes, but it needs  $b$ , which only Alice's wallet holds for the instant of registration; the registration NIZK proves it without revealing it.

## 5.2 Generate Fresh Identity Key Pair and ElGamal Encryption

Alice generates a new key pair on `alt_bn128` and encrypts her identity scalar  $m$  under her own public key using ElGamal:

$$E_{\text{alice}} = (r \cdot G, m \cdot G + r \cdot pk_{\text{alice}})$$

**What this proves:** Alice can encrypt her identity under her own public key and recover it. ElGamal decryption yields the *point*  $M = m \cdot G$ , not the scalar  $m$  (solving the discrete log is infeasible).

```

# Fresh identity key pair for this account
sk_alice = rand()
pk_alice = mul(G1, sk_alice)

# ElGamal encryption of m under pk_alice
r_enc = rand()
R_alice = mul(G1, r_enc)
M_alice = mul(G1, m_alice)
C_alice = add(M_alice, mul(pk_alice, r_enc))
E_alice = (R_alice, C_alice)

# R = r * G
# M = m * G (message point)
# C = m*G + r*pk

print(f"Fresh key pair for this Ethereum account:")
print(f"  sk_alice = {sk_alice}")
print(f"  pk_alice = {pt(pk_alice)}")
print(f"\nElGamal ciphertext E_alice = (R, C):")
print(f"  R = r*G      = {pt(R_alice)}")
print(f"  C = m*G+r*pk = {pt(C_alice)}")
print(f"\nMessage point M = m*G:")
print(f"  M = {pt(M_alice)}")

# Verify Alice can decrypt her own ciphertext
M_check = add(C_alice, neg(mul(R_alice, sk_alice))) # C - sk*R = M
assert eq(M_check, M_alice), "ElGamal self-decryption FAILED"
print(f"\nAlice decrypts: C - sk*R = {pt(M_check)} (matches M)")

```

**How to read the output:** The last two lines show the decrypted point  $C - sk \cdot R$  and the original  $M$ . Their coordinate fragments must match – this confirms the ciphertext is well-formed. If Alice used the wrong secret key, the decrypted point would be a random, unrelated point (shown in the counter-example below).

### 5.2.1 Counter-example: Wrong Secret Key

An attacker who doesn't know `sk_alice` cannot decrypt to the correct  $M$ . The result is an unrelated random point:

```

sk_eve = rand() # Eve's key, not Alice's
M_wrong = add(C_alice, neg(mul(R_alice, sk_eve)))
print(f"Eve decrypts with wrong key:")
print(f"  M_wrong = {pt(M_wrong)}")
print(f"  M_real  = {pt(M_alice)}")
print(f"  Match: {eq(M_wrong, M_alice)}")
print(f"  (Completely different point -- no information about m is leaked.)")

```

### 5.3 NIZK Proof: Binding the Presentation to the ElGamal Ciphertext

Alice must prove, without revealing  $m$ ,  $b$ ,  $r$  or  $sk$ , that:

- (a')  $(A, B)$  presents a valid PS credential on  $m$ :  $e(B, g_2) = e(A, X) \cdot e(mA + bG, Y)$
- (b)  $E_{\text{alice}}$  encrypts the same  $m$
- (k) she holds  $sk$  for  $pk$

This is a Schnorr-family sigma protocol with four nonces  $(\tilde{m}, \tilde{b}, \tilde{r}, \tilde{sk})$  and four commitments:  $C_1 = \tilde{m}A + \tilde{b}G$  (one commitment for both credential exponents, so no proof field ever carries  $\tilde{m}A$  alone),  $T_C = \tilde{m}G + \tilde{r}pk$ ,  $T_R = \tilde{r}G$  and  $T_{key} = \tilde{sk}G$ ; the shared  $\tilde{m}$  is what joins the credential to the ciphertext. The Fiat-Shamir challenge binds the registrant address, the chain id, the registry address and the domain `AlbertaBuck/FiatShamir/IdentityRegistry/Register/v3`, so a proof cannot be replayed to another account, chain, registry or protocol version. The wallet's `registration_prove` is the executable specification – the Solidity, Rust and JavaScript verifiers are checked against its vectors – and it is the single registration path in this document.

```

from alberta_buck.wallet import ElGamalCiphertext, registration_prove

E_alice_w = ElGamalCiphertext(R=R_alice, C=C_alice)
proof_w = registration_prove(
    pres_alice, b_alice, cred.m, r_enc, pk_alice, E_alice_w,
    ALICE_ADDR, sk_alice, rng=rand,
)
print("Registration proof pi = (e, s_m, s_b, s_r, s_sk, C1, T_C, T_R, T_key)")
print(f"  e   = {proof_w.e}")
print(f"  s_m = {proof_w.s_m}")
print(f"  s_b = {proof_w.s_b}")
print(f"  C1  = {pt(proof_w.C1)}  (m~*A + b~*G -- never m~*A on its own)")

```

## 6 Registration – On-Chain Verification

Alice submits  $(\text{issuer}, pk, E_{\text{alice}}, (A, B), \pi)$  to the IdentityRegistry contract. The contract runs six checks using the `ecPairing`, `ecMul` and `ecAdd` precompiles (443,892 gas measured on the A' branch with an in-process EVM, against 472,630 for the pre-A' verifier): a guard on the points and scalars, a Fiat-Shamir reconstruction, two ElGamal consistency checks, the account-key check, and one three-pair pairing product. All six must pass for `isVerified[alice]` to be set. The full verification runs below, followed by two counter-examples: a ciphertext containing a different  $m$  than the presented credential, and a replay of the proof to another address.

### 6.1 What the Chain Sees, and What It Can Learn

Calldata is permanent and visible to every node, so the security of the identity layer rests on one principle: nothing submitted to the EVM is identity data, or a value that anyone – the issuer included – can test against identity data. The plaintext `identity_data`, the scalar  $m$ , the point

$M = mG$ , the raw credential  $\sigma$ , the account secret  $sk$  and the blinding  $b$  never leave the wallet. Each of the five arguments of **register** is a public key, a ciphertext, a blinded pair or a masked scalar, and the contract only checks relations among them with pairings and scalar multiplications; it holds no key and decrypts nothing.

| Argument                                                  | What it is                                                           | What the EVM (and every archive node) can learn from it                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>issuer</b>                                             | The issuer's Ethereum address                                        | Which trust anchor certified the holder. Public by design; it narrows the anonymity set to that issuer's clients, nothing more.                                                                                                                                                                                                                                                    |
| <b>pk</b>                                                 | A fresh account key $sk \cdot G$                                     | Nothing about the identity: it is independent of $m$ .                                                                                                                                                                                                                                                                                                                             |
| $E_{\text{alice}} = (rG, mG + r \cdot pk)$                | ElGamal ciphertext of the identity point under the account's own key | The plaintext needs $sk$ , never sent. Testing whether it encrypts a candidate $M_i$ is the DDH plaintext-checking problem.                                                                                                                                                                                                                                                        |
| $(A, B) = (a\sigma_1, a\sigma_2 + bY_1)$                  | The blinded presentation of the credential                           | A uniform pair of points, the same distribution for every identity.                                                                                                                                                                                                                                                                                                                |
| $pi = (e, s_m, s_b, s_r, s_{sk}, C_1, T_C, T_R, T_{key})$ | The zero-knowledge proof                                             | Not a signature anyone can verify; $\sigma$ itself is never published. Each response is a secret one-time nonce plus a multiple of the witness ( $s_m = \tilde{m} + em$ ), so it is uniform whatever $m$ is; the commitments are simulatable from public values; the only credential-side point derivable is $P = mA + bG$ , which every candidate identity explains equally well. |

Two distinctions matter. The pre-A' design also kept **identity\_data** off chain, but it published a rerandomized signature and a commitment  $\tilde{m}\sigma'_1$  that were *testable* against a known  $m$ ; for anyone holding a candidate list, above all the issuer, a testable function of the identity is as revealing as the identity itself (review findings R1 and R2). The presentation and the single commitment  $C_1$  remove that test; the appendix runs it both ways. And the guarantees differ in kind: the presentation and the proof hide  $m$  information-theoretically, while the ciphertext hides  $M$  only computationally, under DDH – the harvest-now-decrypt-later caveat the papers carry. The hiding of  $s_m$  also depends on the nonce  $\tilde{m}$  being fresh for every proof, which the wallet's sampler is responsible for; two proofs sharing a nonce would open  $m$  by one subtraction.

## 6.2 Verify the NIZK Proof (What the Contract Computes)

**What this proves:** the same  $m$  is presented in  $(A, B)$  and encrypted in  $E_{\text{alice}}$ , and the registrant holds  $sk$  – without revealing  $m$ ,  $b$ ,  $r$  or  $sk$ . Six checks run:

- **(g)** Guards:  $A \neq O$ ,  $B \neq O$ ,  $pk \neq O$ ,  $R \neq O$ ; all points on the curve; scalars canonical.  $A \neq O$  is security critical: at  $A = O$  the pairing product below holds for every  $m$ .
- **(fs)** Fiat-Shamir: the challenge was honestly derived over  $(A, B, R, C, pk, C_1, T_C, T_R, T_{key})$ , the registrant, the chain id, the registry and the domain
- **(b)** ElGamal C consistency: the committed  $m$  is inside  $C$
- **(c)** ElGamal R consistency: the committed  $r$  is inside  $R$
- **(k)** Account key: the committed  $sk$  is the logarithm of  $pk$
- **(a')** Pairing product:  $e(s_m A + s_b G - C_1, Y) \cdot e(eA, X) \cdot e(-eB, g_2) = 1$

```
from alberta_buck.wallet.bn254 import point_to_words
from alberta_buck.wallet.nizk import REGISTER_DOMAIN
from alberta_buck.wallet.transcript import keccak_scalar

# === VERIFIER (IdentityRegistry contract, on-chain) ===
e_v, s_m_v, s_b_v, s_r_v, s_sk_v = (proof_w.e, proof_w.s_m, proof_w.s_b,
                                     proof_w.s_r, proof_w.s_sk)
C1_v, T_C_v, T_R_v, T_key_v = proof_w.C1, proof_w.T_C, proof_w.T_R, proof_w.T_key
CHAINID, REGISTRY = 1, 0      # the wallet's defaults; the contract uses its own
```

```

# --- (g) guards ---
assert not any(is_inf(P) for P in (A_alice, B_alice, pk_alice, R_alice))
assert all(0 <= s_ < ORDER for s_ in (e_v, s_m_v, s_b_v, s_r_v, s_sk_v))
print("(g) guards passed: A, B, pk, R != 0; scalars canonical")

# --- (fs) Fiat-Shamir over the same words the contract hashes ---
words = []
for P in (A_alice, B_alice, R_alice, C_alice, pk_alice, C1_v, T_C_v, T_R_v, T_key_v):
    words += list(point_to_words(P))
e_recomputed = keccak_scalar(*words, ALICE_ADDR, CHAINID, REGISTRY, REGISTER_DOMAIN)
assert e_recomputed == e_v, "Fiat-Shamir challenge mismatch"
print("(fs) Fiat-Shamir challenge verified (A, B, E, pk, commitments, registrant, chain, registry, domain)")

# --- (b) ElGamal C consistency: s_m*G + s_r*pk == e*C + T_C ---
assert eq(add(mul(G1, s_m_v), mul(pk_alice, s_r_v)), add(mul(C_alice, e_v), T_C_v))
print("(b) ElGamal C check passed: s_m*G + s_r*pk == e*C + T_C")

# --- (c) ElGamal R consistency: s_r*G == e*R + T_R ---
assert eq(mul(G1, s_r_v), add(mul(R_alice, e_v), T_R_v))
print("(c) ElGamal R check passed: s_r*G == e*R + T_R")

# --- (k) account key: s_sk*G == T_key + e*pk ---
assert eq(mul(G1, s_sk_v), add(T_key_v, mul(pk_alice, e_v)))
print("(k) account-key check passed: s_sk*G == T_key + e*pk")

# --- (a') pairing product (what ecPairing computes: three pairs) ---
lhs = add(add(mul(A_alice, s_m_v), mul(G1, s_b_v)), neg(C1_v)) # = e*(m*A + b*G)
check = (
    pairing(IPK_Y, lhs) # e(s_m*A + s_b*G - C1, Y)
    * pairing(IPK_X, mul(A_alice, e_v)) # e(e*A, X)
    * pairing(G2, neg(mul(B_alice, e_v))) # e(-e*B, g_2)
)
assert check == FQ12_one(), "presentation pairing check FAILED"
print("(a') pairing product passed:")
print("    e(s_m*A + s_b*G - C1, Y) * e(e*A, X) * e(-e*B, g_2) == 1")

# The only credential-side point anyone can derive from the proof:
P_derived = mul(lhs, pow(e_v, -1, ORDER))
assert eq(P_derived, P_alice)
print("    derivable point P = m*A + b*G: uniform, and the same for every candidate m")
print("\n=> Registration PASSED. isVerified(alice) = true")

```

**How to read the output:** All six checks must print "passed". If any assertion fires, the proof is invalid. The last line before the verdict names the one credential-side value an observer can compute from the proof:  $P = mA + bG$ , which every candidate  $m_i$  explains equally well (as  $m_iA + (P - m_iA)$ ), so the proof fields support no test on the identity. The pre-A' proof published  $\tilde{m}A$  as a separate commitment, from which  $mA$  followed by one subtraction.

### 6.2.1 Counter-example: ElGamal Encrypts a Different $m$

If Alice (or an attacker) encrypts one  $m$  in ElGamal but presents a credential on a *different*  $m$ , the shared nonce  $\tilde{m}$  forces the two relations apart and the ElGamal consistency check fails. The wallet's `registration_verify` runs the same six checks as the contract (on the compiled kernel), and also rejects a replay of the honest proof to another registrant.

```

from alberta_buck.wallet import registration_verify

# Forge: encrypt a fake m in ElGamal, but present the real credential
r_fake = rand()
E_fake = ElGamalCiphertext(R=mul(G1, r_fake),
                           C=add(mul(G1, m_fake), mul(pk_alice, r_fake)))
proof_f = registration_prove(pres_alice, b_alice, cred.m, r_fake, pk_alice,

```

```

        E_fake, ALICE_ADDR, sk_alice, rng=rand)
ok_honest = registration_verify(pres_alice, E_alice_w, pk_alice,
                               issuer_obj.pk_X, issuer_obj.pk_Y, proof_w, ALICE_ADDR)
ok_forged = registration_verify(pres_alice, E_fake, pk_alice,
                               issuer_obj.pk_X, issuer_obj.pk_Y, proof_f, ALICE_ADDR)
# Replay of the honest proof to a different registrant address
ok_replay = registration_verify(pres_alice, E_alice_w, pk_alice,
                               issuer_obj.pk_X, issuer_obj.pk_Y, proof_w, ALICE_ADDR ^ 1)
print(f"registration_verify(honest E_alice)           : {ok_honest}")
print(f"registration_verify(E_fake encrypts m_fake)    : {ok_forged}")
print(f"registration_verify(replayed to another address): {ok_replay}")

```

## 7 Bob Binds His Pool as a Public-Identity Contract

Bob operates a Uniswap-style AMM liquidity pool. Bob himself is a registered EOA, but the *pool contract* is the entity that holds and moves BUCK on his behalf. The governance-approved adapter for the particular Uniswap factory checks that Bob is its registered `feeToSetter` (V2) or `owner` (V3), creates or locates the canonical pool, and calls `IdentityRegistry.bindContractFromAdapter` atomically, publishing Bob's (`pk`, `E`) against the pool's address and marking the pool `isPublicIdentity = true`. Bob's `identity_data` remains world-readable in the `IdentityRegistry` under his own EOA.

Counterparties of the pool need not wait for the pool to call `approve` back: when no per-pair receipt fragment exists for the pool side, `transfer` falls back to the bound `_identityHash(pool)` for the receipt event. The bilateral identity check still requires both parties to be `isVerified`; the public-side fallback only relaxes the receipt-fragment requirement, not the verification requirement.

```

# Bob's identity (public pool operator)
bob_identity_data = json.dumps({
    "given_name": "Bob",
    "family_name": "Smith",
    "jurisdiction": "Alberta, Canada",
    "id_type": "Corporate Registration",
    "id_number": "AB-CORP-2026-00182",
    "date_of_birth": "1985-07-22",
    "issuer_id": "atb-financial-ca",
    "issued_at": "2026-02-01T09:00:00Z",
    "epoch": 42
}, sort_keys=True, separators=(',', ':'))

m_bob = keccak(bob_identity_data)

# Bob also gets a PS signature (same issuance ceremony)
t_h_bob = rand()
h_bob = mul(G1, t_h_bob)
sigma_bob = (h_bob, mul(h_bob, (isk_x + m_bob * isk_y) % ORDER))

# Bob generates key pair and ElGamal encryption
sk_bob = rand()
pk_bob = mul(G1, sk_bob)
r_bob = rand()
R_bob = mul(G1, r_bob)
M_bob = mul(G1, m_bob)
C_bob = add(M_bob, mul(pk_bob, r_bob))
E_bob_reg = (R_bob, C_bob)

# Bob registers as an EOA (same PS+NIZK verification as Alice -- omitted for brevity)
# His pool is then bound under a Public Identity through the audited
# adapter for its canonical Uniswap factory.
POOL_ADDR = 0xa11ce0fa11ce0fa11ce0fa11ce0fa1100 # illustrative
print("Bob registered (EOA), pool bound as PUBLIC-IDENTITY contract:")
print(f"  pk_bob = {pt(pk_bob)}")
print(f"  E_bob = (R, C) -- ElGamal ciphertext on-chain")
print(f"  identity_data = '{bob_identity_data[:50]}...'")

```

```
print(f"    isVerified(bob)           = true")
print(f"    isVerified(pool)         = true    (set by bindContract)")
print(f"    isPublicIdentity(pool)     = true    (set by bindContract)")
print(f"\n\nThe pool's identity hash falls back into receipts when no per-pair")
print(f"fragment exists.  Bob himself never needs to approve his own pool.")
```

## 7.1 Same Operation via the Wallet API

Bob's KYC is the identical ceremony as Alice's, just with corporate fields and a different applicant address. `Issuer.issue(...)` produces a fresh sigma each call (random `h` per the PS spec) but verifies under the same issuer key.

```
BOB_ADDR = 0x0b0b00000000000000000000000000000000000000000b0b

bob_fields = {
    "given_name":    "Bob",
    "family_name":   "Smith",
    "jurisdiction":  "Alberta, Canada",
    "id_type":        "Corporate Registration",
    "id_number":      "AB-CORP-2026-00182",
    "date_of_birth":  "1985-07-22",
    "issuer_id":      "atb-financial-ca",
    "issued_at":      "2026-02-01T09:00:00Z",
    "epoch":          42,
}

cred_bob = issuer_obj.issue(bob_fields, BOB_ADDR, rng=rand)
assert cred_bob.m == m_bob
assert ps_verify(issuer_obj.pk_X, issuer_obj.pk_Y, cred_bob.sigma, cred_bob.m)
print(f"Wallet API: cred_bob.m == m_bob      : {cred_bob.m == m_bob}")
print(f"    ps_verify(cred_bob.sigma, m_bob) : True")
print(f"    Issuer log now has 2 entries: {len(issuer_obj.issuance_log())}")
```

## 8 Alice Approves Bob – Re-Encryption + Chaum-Pedersen Proof

Alice wants to swap BUCK for USDT on Bob's pool. She calls `approve(pool, amount, E_pool, proof)` to re-encrypt her identity under the pool's bound public key (Bob's `pk_bob`) and prove the re-encryption is correct via a Chaum-Pedersen proof. Approve always requires CP, even though the pool is a Public-Identity contract – this captures Alice's per-pair consent, not just the registry-derivable identity hash; see *Why approve Is Always CP-Bound* for the full rationale. The pool itself need not reciprocate with its own `approve`; the bilateral identity check in `transfer` passes both guards because `isPublicIdentity(pool) == true` satisfies the public-side fallback, and the receipt carries Alice's per-pair CP fragment alongside the deterministic identity hashes. Below: the re-encryption, the four-check Chaum-Pedersen verification, a counter-example showing that re-encrypting a *different* message point causes Check 3 to fail, and the false-identity forgery that only the key-binding Check 1 stops.

### 8.1 Re-Encrypt Alice's Identity for Bob

Alice decrypts her own ciphertext to recover  $M = m \cdot G$ , then re-encrypts under Bob's public key with fresh randomness.

**What this proves:** Re-encryption preserves the message point  $M$ . Bob can decrypt the new ciphertext with his own secret key and recover the same  $M$  that was in Alice's original ciphertext.

```
# Alice decrypts her own credential
M_decrypted = add(C_alice, neg(mul(R_alice, sk_alice))) #  $C - sk \cdot R = M$ 
assert eq(M_decrypted, M_alice), "Self-decryption failed"
```

```

# Re-encrypt M under Bob's public key with fresh randomness r'
r_prime = rand()
R_for_bob = mul(G1, r_prime)          # R' = r' * G
C_for_bob = add(M_alice, mul(pk_bob, r_prime)) # C' = M + r' * pk_bob
E_for_bob = (R_for_bob, C_for_bob)

print("Alice re-encrypts her identity for Bob:")
print(f"  E_bob = (R', C') where:")
print(f"    R' = r'*G      = {pt(R_for_bob)}")
print(f"    C' = M+r'*pk    = {pt(C_for_bob)}")

# Verify Bob can decrypt
M_bob_decrypts = add(C_for_bob, neg(mul(R_for_bob, sk_bob)))
assert eq(M_bob_decrypts, M_alice), "Bob cannot decrypt E_bob"
print(f"\nBob decrypts: C' - sk_bob*R' = {pt(M_bob_decrypts)}")
print(f"Matches Alice's M:          {pt(M_alice)}")

```

**How to read the output:** The last two lines show Bob's decrypted point and Alice's original  $M$ . The coordinate fragments must be identical – same point, different ciphertext wrapping it.

## 8.2 Chaum-Pedersen Proof of Correct Re-Encryption

Alice proves that  $E_{\text{bob}}$  encrypts the same message point  $M$  as her registered  $E_{\text{alice}}$  – without revealing  $m$ ,  $sk_{\text{alice}}$  or the re-encryption randomness  $r'$ . The statement has three relations, and all three are load-bearing:

$$pk_{\text{alice}} = sk \cdot G, \quad R_{\text{bob}} = r' \cdot G, \quad C_{\text{alice}} - C_{\text{bob}} = sk \cdot R_{\text{alice}} - r' \cdot pk_{\text{bob}}.$$

The first binds the witness to Alice's *registered* key. Without it (the two-relation form an earlier version of this document showed) a sender who knows her own  $sk$  and  $r$  can choose a fake  $sk^*$  that makes  $C_{\text{alice}} - sk^* R_{\text{alice}}$  equal a *third party's* identity point, and hand Bob a ciphertext of that identity with an accepting proof (review finding 3). The wallet's `chaum_pedersen_prove` is the executable specification; its transcript binds (`sender`, `spender`, `chainid`, `registry`).

```

from alberta_buck.wallet import chaum_pedersen_prove, chaum_pedersen_verify

E_for_bob_w = ElGamalCiphertext(R=R_for_bob, C=C_for_bob)
proof_cp = chaum_pedersen_prove(
    E_alice_w, E_for_bob_w, pk_alice, pk_bob,
    sk_alice, r_prime, ALICE_ADDR, BOB_ADDR, 1, rng=rand,
)
print("Chaum-Pedersen proof (e, s1, s2, T1, T2, T3):")
print(f"  e = {proof_cp.e}")
print(f"  s1 = {proof_cp.s1}    (= a + e*sk)")
print(f"  s2 = {proof_cp.s2}    (= b + e*r')")

```

## 8.3 Verify Chaum-Pedersen Proof (What the Contract Computes)

The BUCK contract reads  $E_{\text{alice}}$  and  $pk_{\text{alice}}$  from the IdentityRegistry (never from calldata), then verifies the proof. (Its gas is not re-measured on this branch; the identity paper's figure of roughly 29K predates the three-relation repair.)

**What this proves:**  $E_{\text{alice}}$  and  $E_{\text{bob}}$  encrypt the same  $M$ , and the prover holds the key registered as  $pk_{\text{alice}}$ . Four checks:

- **Check 1:**  $s_1 G = T_1 + e pk_{\text{alice}}$  – the witness is Alice's registered key
- **Check 2:**  $s_2 G = T_3 + e R_{\text{bob}} - r'$  is consistent with  $R_{\text{bob}}$

- **Check 3:**  $s_1 R_{\text{alice}} - s_2 p k_{\text{bob}} = T_2 + e(C_{\text{alice}} - C_{\text{bob}}) - \text{same } M$
- **Check 4:** the Fiat-Shamir challenge was honestly derived

```
# === VERIFIER (BUCK contract, on-chain) ===
e_v, s1_v, s2_v = proof_cp.e, proof_cp.s1, proof_cp.s2
T1, T2, T3 = proof_cp.T1, proof_cp.T2, proof_cp.T3

assert eq(mul(G1, s1_v), add(T1, mul(pk_alice, e_v))), "check 1 FAILED"
print("Check 1 passed: s1*G == T1 + e*pk_alice      (witness is the registered key)")
assert eq(mul(G1, s2_v), add(T3, mul(R_for_bob, e_v))), "check 2 FAILED"
print("Check 2 passed: s2*G == T3 + e*R_bob        (r' consistent with R_bob)")
lhs = add(mul(R_alice, s1_v), neg(mul(pk_bob, s2_v)))
rhs = add(T2, mul(add(C_alice, neg(C_for_bob)), e_v))
assert eq(lhs, rhs), "check 3 FAILED"
print("Check 3 passed: s1*R_a - s2*pk_b == T2 + e*(C_a - C_b)    (same M)")
ok = chaum_pedersen_verify(E_alice_w, E_for_bob_w, pk_alice, pk_bob,
                           proof_cp, ALICE_ADDR, BOB_ADDR, 1)

assert ok
print(f"Check 4 passed: Fiat-Shamir challenge verified (wallet verifier agrees: {ok})")
print("\n==> approve() PASSED. Alice's re-encrypted identity stored for Bob.")
print("      _receiptFragments[alice][bob] = keccak256(E_bob)")
```

**How to read the output:** All four "passed" lines must appear. The two counter-examples below each break exactly one of them.

### 8.3.1 Counter-example: Re-Encryption with Wrong Message

If Alice tries to pass off a different identity point in the re-encrypted ciphertext, Check 3 catches the discrepancy:

```
# Alice tries to re-encrypt a DIFFERENT M (e.g., someone else's identity)
m_other = rand()
M_other = mul(G1, m_other)
r_bad = rand()
R_bad = mul(G1, r_bad)
C_bad = add(M_other, mul(pk_bob, r_bad)) # encrypts M_other, not M_alice
E_bad_w = ElGamalCiphertext(R=R_bad, C=C_bad)
proof_bad = chaum_pedersen_prove(E_alice_w, E_bad_w, pk_alice, pk_bob,
                                 sk_alice, r_bad, ALICE_ADDR, BOB_ADDR, 1, rng=rand)
lhs = add(mul(R_alice, proof_bad.s1), neg(mul(pk_bob, proof_bad.s2)))
rhs = add(proof_bad.T2, mul(add(C_alice, neg(C_bad)), proof_bad.e))
print(f"Forged re-encryption -- Check 3 (same M) : {eq(lhs, rhs)}")
ok_bad = chaum_pedersen_verify(E_alice_w, E_bad_w, pk_alice, pk_bob,
                               proof_bad, ALICE_ADDR, BOB_ADDR, 1)
print(f"Forged re-encryption -- wallet verifier : {ok_bad}")
print(" (FAILS: C_bad encrypts a different M than C_alice; approve() reverts.)")
```

### 8.3.2 Counter-example: A Third Party's Identity Under a Fake Key

The forgery the three-relation statement exists to stop. Alice knows her own  $sk$  and  $r$ , so she can solve for a fake key  $sk^*$  with  $C_{\text{alice}} - sk^* R_{\text{alice}} = M_{\text{victim}}$  and encrypt the victim's point for Bob. Checks 2 and 3 pass – the prover is honest about *its* witnesses – and only Check 1, the binding to her registered key, rejects it. The counter-example is the review suite's own (alberta\_buck.review.examples).

```
from alberta_buck.review.examples import false_identity_approval
from alberta_buck.wallet import elgamal_decrypt

alice_r, bob_r, victim_m, fake_sk, r_star, forged, proof_x = false_identity_approval(
    ALICE_ADDR, BOB_ADDR, 1)
```

```

# Bob would decrypt the forged ciphertext to the VICTIM's identity point
print(f"forged E decrypts (under sk_bob) to victim's M : "
      f"{eq(elgamal_decrypt(forged, bob_r.sk), mul(G1, victim_m))}")
print(f"fake witness sk* is Alice's registered key      : {eq(mul(G1, fake_sk), alice_r.pk)}")
c2 = eq(mul(G1, proof_x.s2), add(proof_x.T3, mul(forged.R, proof_x.e)))
c3 = eq(add(mul(alice_r.E.R, proof_x.s1), neg(mul(bob_r.pk, proof_x.s2))),
        add(proof_x.T2, mul(add(alice_r.E.C, neg(forged.C)), proof_x.e)))
c1 = eq(mul(G1, proof_x.s1), add(proof_x.T1, mul(alice_r.pk, proof_x.e)))
print(f"Checks 2 and 3 (the two-relation protocol)      : {c2 and c3}")
print(f"Check 1 (witness == registered key)             : {c1}")
ok_x = chaum_pedersen_verify(alice_r.E, forged, alice_r.pk, bob_r.pk, proof_x,
                             ALICE_ADDR, BOB_ADDR, 1)
print(f"wallet verifier (three relations)              : {ok_x}")

forged E decrypts (under sk_bob) to victim's M : True
fake witness sk* is Alice's registered key      : False
Checks 2 and 3 (the two-relation protocol)      : True
Check 1 (witness == registered key)             : False
wallet verifier (three relations)              : False

```

## 9 Transfer – Bilateral Identity Check

With Alice's approve completed and the pool bound under a Public Identity, the **transfer** can proceed. The contract performs a bilateral identity check: both parties must be **isVerified**, and the receipt event must carry a fragment for each side. Per-pair receipt fragments stored at **approve** time are the primary source. When a side has no fragment and that side is **isPublicIdentity = true**, the bound **\_identityHash** falls in for that side's receipt slot. No cryptography runs at transfer time – only state lookups. We simulate the check for our Private-to-Public-Identity scenario.

```

# Simulating the contract's bilateral identity check
# Alice (private EOA) -> Pool (public contract)
is_verified_alice      = True
is_verified_pool       = True
is_public_alice        = False # EOA, not public-identity
is_public_pool         = True  # bindContract(pool, ..., true)
receipt_alice_to_pool  = True  # set during approve() above
receipt_pool_to_alice  = False # pool never CP-approved alice

assert is_verified_alice and is_verified_pool

# toHash = _receiptFragments[from][to] -- Alice->Pool direction
if receipt_alice_to_pool:
    toHash = "H(E_pool)" # Alice's per-pair CP fragment
elif is_public_alice or is_public_pool:
    toHash = "_identityHash(pool)" # fallback (pool is public)
else:
    raise Exception("sender must identity-approve recipient")
# fromHash = _receiptFragments[to][from] -- Pool->Alice direction
if receipt_pool_to_alice:
    fromHash = "H(E_alice)" # Pool's CP fragment (doesn't exist)
elif is_public_alice or is_public_pool:
    fromHash = "_identityHash(alice)" # fallback (pool is public)
else:
    raise Exception("recipient must identity-approve sender")

print("Bilateral identity check:")
print(f"  toHash (Alice->Pool): {toHash}")
print(f"  fromHash (Pool->Alice): {fromHash}")
print(f"\n  both hashes present => transfer SUCCEEDS")
print(f"\n=> transfer(alice, pool, 500 BUCK) SUCCEEDS")
print(f"  BuckTransferReceipt(fromHash={fromHash}, toHash={toHash})")

```

```

Bilateral identity check:
  toHash (Alice->Pool): H(E_pool)
  fromHash (Pool->Alice): _identityHash(alice)

both hashes present => transfer SUCCEEDS

==> transfer(alice, pool, 500 BUCK) SUCCEEDS
      BuckTransferReceipt(fromHash=_identityHash(alice), toHash=H(E_pool))

```

## 10 Bob Verifies Alice's Identity (Two-Channel Design)

This is the two-channel design in action. Channel 1 (Ethereum): Bob decrypts  $E_{\text{pool}}$  with his secret key to recover the message point  $M$ . Channel 2 (Holochain): Bob receives Alice's `identity_data` and verifies the binding  $H(\text{identity\_data}) \cdot G = M$ . Neither channel alone is sufficient – the Ethereum ciphertext proves *which* identity was registered, while the Holochain data provides the *human-readable content*. A counter-example shows that changing even one character of `identity_data` in transit (e.g., "Alice" to "Mallory") produces a completely different point that doesn't match  $M$ , so Bob detects tampering immediately.

```

# === Bob's side ===

# Channel 1 (Ethereum proof chain): Bob decrypts E_bob
M_from_bob = add(C_for_bob, neg(mul(R_for_bob, sk_bob)))
print("Bob decrypts E_bob from on-chain ApproveReceipt event:")
print(f"  M = C' - sk_bob * R' = {pt(M_from_bob)}")

# Channel 2 (Holochain data channel): Bob receives identity_data
received_identity_data = alice_identity_data # <-- via Holochain, encrypted for Bob
print(f"Bob receives identity_data via Holochain:")
print(f"  '{received_identity_data[:60]}...'")

# Verify binding: H(identity_data) * G == M
m_check = keccak(received_identity_data)
M_check = mul(G1, m_check)
assert eq(M_check, M_from_bob), "Identity binding FAILED"

print(f"\nTwo-channel verification:")
print(f"  m = H(identity_data) = {m_check}")
print(f"  m * G = {pt(M_check)}")
print(f"  M      = {pt(M_from_bob)}")
print(f"  H(identity_data) * G == M : True")
print(f"\n==> Bob has cryptographic proof that the identity_data")
print(f"      he received is the same identity the issuer certified")
print(f"      and Alice registered on-chain.")

```

**How to read the output:** The coordinate fragments of  $m \cdot G$  and  $M$  (decrypted from on-chain) must be identical. A counter-example follows showing what Bob sees if the Holochain data is tampered.

### 10.1 Counter-example: Tampered Identity Data

If a man-in-the-middle alters the identity data in transit, Bob detects it immediately – the hash no longer matches the on-chain point:

```

# Someone tampers with the identity data in transit
tampered = alice_identity_data.replace("Alice", "Mallory")
m_tampered = keccak(tampered)
M_tampered = mul(G1, m_tampered)

print(f"Tampered identity_data: ...given_name: 'Mallory'...")

```

```

print(f"  H(tampered) * G = {pt(M_tampered)}")
print(f"  M (from chain) = {pt(M_from_bob)}")
print(f"  Match: {eq(M_tampered, M_from_bob)}")
print(f"  (Bob rejects: the off-chain data doesn't match the on-chain point.)")

```

## 11 Receipt Construction

After the transfer, both parties hold each other's plaintext `identity_data` (Alice has Bob's from the public IdentityRegistry; Bob has Alice's from Holochain) and the transfer metadata from on-chain events. Either party can independently construct a canonical receipt – deterministic JSON with sorted keys and no whitespace – and compute its `keccak256` hash. Alice's and Bob's independently constructed receipts produce identical hashes without coordination. This canonical form is what makes dispute resolution possible: both parties hold the same evidence, and neither can alter their copy without breaking the hash.

```

# Both parties construct the same canonical receipt
def canonical_receipt(sender_id, receiver_id, amount, tx_hash, block):
    return json.dumps({
        "sender": json.loads(sender_id),
        "receiver": json.loads(receiver_id),
        "amount": amount,
        "tx_hash": tx_hash,
        "block": block,
    }, sort_keys=True, separators=(',', ':'))

# Stand-in transaction data (would come from on-chain Transfer event)
tx_hash = "0xabcdef1234567890"
block = 19400000
amount = 500

# Alice constructs her copy
receipt_alice = canonical_receipt(
    alice_identity_data, bob_identity_data, amount, tx_hash, block)

# Bob constructs his copy
receipt_bob = canonical_receipt(
    alice_identity_data, bob_identity_data, amount, tx_hash, block)

h_alice = keccak(receipt_alice)
h_bob = keccak(receipt_bob)

print("Receipt construction (independent, canonical):")
print(f"  Alice's H(receipt) = {h_alice}")
print(f"  Bob's H(receipt) = {h_bob}")
print(f"  Match: {h_alice == h_bob}")
print(f"\nReceipt stored (encrypted) on each party's Holochain source chain.")
print(f"The receipt hash anchors the off-chain record to the on-chain transfer.")

```

## 12 Summary: What Each Party Knows

After a complete Private-to-Public transfer, the information landscape is sharply partitioned. Eve (a passive on-chain observer) learns nothing about Alice's identity from any public artifact. Bob learns Alice's identity only because Alice explicitly authorized it via `approve`. The issuer, and Bob once he has learned  $m$ , hold a candidate they could test against public transcripts – and the presentation  $(A, B)$  is built so that the test fails: it is a uniform pair, the same distribution for every identity, and the only point derivable from its proof is explained by every candidate. What remains is the ElGamal ciphertext, whose lock is computational (DDH), and the credential itself,

which Bob never sees: knowing  $m$  does not let him register under it (that needs  $\sigma$  and a fresh  $b$ ), only the issuer can certify it again.

```
[
  ["Artifact", "Alice", "Bob", "Eve", "Issuer"],
  None,
  ["identity_data (Alice)", "YES (owns it)", "YES (Holochain)", "NO", "YES (issued it)"],
  ["identity_data (Bob)", "YES (public)", "YES (owns it)", "YES (public)", "YES (issued it)"],
  ["m = H(identity_data)", "YES", "YES (computes)", "NO", "YES"],
  ["M = m*G", "YES", "YES (decrypts)", "NO", "YES (computes)"],
  ["E_alice (registered)", "YES", "sees on-chain", "sees on-chain", "sees on-chain"],
  ["E_bob (re-encrypted)", "YES (created)", "YES (decrypts)", "sees on-chain", "sees on-chain"],
  ["sk_alice", "YES", "NO", "NO", "NO"],
  ["sk_bob", "NO", "YES", "NO", "NO"],
  ["PS credential sigma", "YES", "NO", "NO", "YES (created)"],
  ["Presentation (A, B)", "YES (created)", "sees on-chain", "sees on-chain", "sees on-chain"],
  ["Blinding b", "YES (then discards)", "NO", "NO", "NO"],
  ["Registration proof", "YES (created)", "sees on-chain", "sees on-chain", "sees on-chain"],
  ["Chaum-Pedersen proof", "YES (created)", "verified on-chain", "sees on-chain", "N/A"],
  ["Full receipt", "YES", "YES", "NO", "NO"],
  ["Test (A, B) against m", "trivially", "NO (fails for every m)", "NO", "NO (fails for every m)"],
  ["Test E against m*G", "trivially", "NO (DDH)", "NO (DDH)", "NO (DDH)"],
  ["Register under m", "YES (holds sigma)", "NO (needs sigma, b)", "NO", "YES (may re-certify)"],
]
```

| Artifact                        | Alice               | Bob                    | Eve           | Issuer                 |
|---------------------------------|---------------------|------------------------|---------------|------------------------|
| identity_data (Alice)           | YES (owns it)       | YES (Holochain)        | NO            | YES (issued it)        |
| identity_data (Bob)             | YES (public)        | YES (owns it)          | YES (public)  | YES (issued it)        |
| $m = H(\text{identity\_data})$  | YES                 | YES (computes)         | NO            | YES                    |
| $M = m*G$                       | YES                 | YES (decrypts)         | NO            | YES (computes)         |
| $E_{\text{alice}}$ (registered) | YES                 | sees on-chain          | sees on-chain | sees on-chain          |
| $E_{\text{bob}}$ (re-encrypted) | YES (created)       | YES (decrypts)         | sees on-chain | sees on-chain          |
| $sk_{\text{alice}}$             | YES                 | NO                     | NO            | NO                     |
| $sk_{\text{bob}}$               | NO                  | YES                    | NO            | NO                     |
| PS credential sigma             | YES                 | NO                     | NO            | YES (created)          |
| Presentation (A, B)             | YES (created)       | sees on-chain          | sees on-chain | sees on-chain          |
| Blinding b                      | YES (then discards) | NO                     | NO            | NO                     |
| Registration proof              | YES (created)       | sees on-chain          | sees on-chain | sees on-chain          |
| Chaum-Pedersen proof            | YES (created)       | verified on-chain      | sees on-chain | N/A                    |
| Full receipt                    | YES                 | YES                    | NO            | NO                     |
| Test (A, B) against m           | trivially           | NO (fails for every m) | NO            | NO (fails for every m) |
| Test E against $m*G$            | trivially           | NO (DDH)               | NO (DDH)      | NO (DDH)               |
| Register under m                | YES (holds sigma)   | NO (needs sigma, b)    | NO            | YES (may re-certify)   |

## 13 Appendix: The Identity Fountain – The Two-Sided Linkability Test

Alice derives a second account from the same credential. The question is not whether the two accounts' public artifacts "look different" (any two random points do) but whether a party holding the identity scalar  $m$  – the issuer, or a past counterparty – can *test* them. The block runs that test both ways. On what the pre-A' protocol published (a rerandomized signature) it identifies both accounts as Alice's. On what the A' protocol publishes (the presentations, and the one point

derivable from their proofs) it identifies nothing, because every candidate explains every account equally well. The remaining lock, on the ElGamal ciphertexts, is computational (DDH), as the identity paper’s privacy-boundary table states; the credential lock is information-theoretic.

```
from alberta_buck.wallet.nizk import presentation_point

# == Alice creates a SECOND account: same credential, fresh (a, b, sk, r) ==
pres_2, a_2, b_2 = ps_present(cred.sigma, issuer_obj.pk_Y1, rng=rand)
sk_2 = rand(); pk_2 = mul(G1, sk_2); r_2 = rand()
E_2 = ElGamalCiphertext(R=mul(G1, r_2), C=add(M_alice, mul(pk_2, r_2)))
proof_2 = registration_prove(pres_2, b_2, cred.m, r_2, pk_2, E_2,
                             ALICE_ADDR ^ 2, sk_2, rng=rand)
assert registration_verify(pres_2, E_2, pk_2, issuer_obj.pk_X, issuer_obj.pk_Y,
                           proof_2, ALICE_ADDR ^ 2)
print("Second account registered (same identity, fresh presentation and keys).\n")

# The candidate list an issuer (or a counterparty who learned m) holds
candidates = {"alice": m_alice, "bob": m_bob, "stranger": m_fake}

# Side 1: what the pre-A' protocol published -- rerandomized signatures
t_2 = rand(); sigma_p2 = (mul(sigma[0], t_2), mul(sigma[1], t_2))
print("Pre-A' publication (rerandomized signatures): raw PS test per candidate")
for name, m_i in candidates.items():
    hit = [raw_test(S[0], S[1], m_i) for S in (sigma_p, sigma_p2)]
    print(f" {name:9s}: account 1 {str(hit[0]):5s} account 2 {str(hit[1]):5s}")

# Side 2: what the A' protocol publishes -- presentations
print("A' publication (presentations): raw PS test per candidate")
for name, m_i in candidates.items():
    hit = [raw_test(p.A, p.B, m_i) for p in (pres_alice, pres_2)]
    print(f" {name:9s}: account 1 {str(hit[0]):5s} account 2 {str(hit[1]):5s}")

# The proof fields: the derivable point P = m*A + b*G satisfies the
# masked equation on its own, and P - m_i*A is a well-formed b_i*G
# for EVERY candidate m_i -- the relation discriminates nothing.
print("A' proof fields: the derivable point P explains every candidate")
for k, (p, pf) in enumerate(((pres_alice, proof_w), (pres_2, proof_2)), 1):
    P = presentation_point(p, pf)
    masked = pairing(G2, p.B) == pairing(IPK_X, p.A) * pairing(IPK_Y, P)
    print(f" account {k}: masked equation with derived P -> {masked}; "
          f"P - m_i*A is a point for all {len(candidates)} candidates")
assert not eq(pres_alice.A, pres_2.A) and not eq(pres_alice.B, pres_2.B)
print("Two presentations of one credential share no point.")
```

**How to read the output:** In the first table every account lights up under Alice’s  $m$  and no other: the pre-A’ pair was a candidate test, which is exactly review finding R1. In the second table no account lights up under any candidate, including the true one. The proof fields then confirm that the point they expose is consistent with every candidate. This is the two-sided witness the A’ plan asks for; the hiding argument behind it is Theorem 2’ of the proofs document, and its independent review is still outstanding.

## 14 Appendix: Small-Group Proofs

The BN254 examples above prove the cryptography at production scale, but 77-digit numbers resist hand-verification. These demonstrations use modulus  $p = 23$  working in an order-11 subgroup where every value fits in two digits.

| EC (additive)           | Modular (multiplicative)           |
|-------------------------|------------------------------------|
| Scalar mult $m \cdot G$ | Exponentiation $g^m \bmod p$       |
| Point addition $P + Q$  | Multiplication $P \cdot Q \bmod p$ |

Schnorr proofs require prime-order groups, so we use  $g = 4$ , which generates the order-11 subgroup (the quadratic residues mod 23). All exponent arithmetic is mod  $q = 11$ .

## 14.1 Setup and ElGamal

ElGamal encryption of a group element  $M = g^m$  under public key  $pk = g^{sk}$  with randomness  $r$ :

$$E = (R, C) = (g^r, M \cdot pk^r) \pmod{p}$$

$$\text{Decrypt: } M = C \cdot (R^{sk})^{-1} \pmod{p}$$

```
import hashlib

p = 23; q = 11; g = 4      # safe prime, prime-order subgroup

def mod_exp(base, exp, mod):
    result = 1
    base = base % mod
    while exp > 0:
        if exp & 1:
            result = (result * base) % mod
        base = (base * base) % mod
        exp >>= 1
    return result

def mod_inv(a, mod):
    def egcd(a, b):
        if a == 0: return b, 0, 1
        g, x, y = egcd(b % a, a)
        return g, y - (b // a) * x, x
    _, x, _ = egcd(a % mod, mod)
    return x % mod

def fs_hash(*args):
    """Fiat-Shamir: H(args...) mod q"""
    h = hashlib.sha256()
    for a in args:
        h.update(str(a).encode())
    return int(h.hexdigest(), 16) % q

# Identity
m = 3; M = mod_exp(g, m, p)

# Keys
sk_a = 5; pk_a = mod_exp(g, sk_a, p)
sk_b = 7; pk_b = mod_exp(g, sk_b, p)

# Encrypt for Alice (r_a=2) and Bob (r_b=9)
r_a = 2; R_a = mod_exp(g, r_a, p)
C_a = (M * mod_exp(pk_a, r_a, p)) % p

r_b = 9; R_b = mod_exp(g, r_b, p)
C_b = (M * mod_exp(pk_b, r_b, p)) % p

# Decrypt both
dec_a = (C_a * mod_inv(mod_exp(R_a, sk_a, p), p)) % p
dec_b = (C_b * mod_inv(mod_exp(R_b, sk_b, p), p)) % p

[
    ["", "m", "M=g^m", "sk", "pk", "r", "R=g^r", "C=M*pk^r", "Decrypted"],
    None,
    ["Alice", m, M, sk_a, pk_a, r_a, R_a, C_a, dec_a],
    ["Bob", m, M, sk_b, pk_b, r_b, R_b, C_b, dec_b],
]
```

|       | m | $M=g^m$ | sk | pk | r | $R=g^r$ | $C=M*pk^r$ | Decrypted |
|-------|---|---------|----|----|---|---------|------------|-----------|
| Alice | 3 | 18      | 5  | 12 | 2 | 16      | 16         | 18        |
| Bob   | 3 | 18      | 7  | 8  | 9 | 13      | 1          | 18        |

Same message  $M = 18$  in both ciphertexts: (16,16) for Alice, (13,1) for Bob. Different randomness makes them unlinkable.

## 14.2 NIZK: Proof of Knowledge (Registration)

At registration, Alice proves she knows the plaintext  $m$  and randomness  $r$  inside her ElGamal ciphertext – without revealing either. (In the full protocol, a Pointcheval-Sanders pairing check additionally binds  $m$  to the issuer's signature; this demonstration isolates the sigma-protocol core.)

Statement:  $R = g^r, \quad C = g^m \cdot pk^r$

Commit:  $T_R = g^{k_r}, \quad T_C = g^{k_m} \cdot pk^{k_r}$

Challenge:  $e = H(g, pk, R, C, T_R, T_C) \bmod q$

Respond:  $s_m = k_m - e \cdot m \pmod{q}, \quad s_r = k_r - e \cdot r \pmod{q}$

Verify 1:  $g^{s_r} \cdot R^e \stackrel{?}{=} T_R$

Verify 2:  $g^{s_m} \cdot pk^{s_r} \cdot C^e \stackrel{?}{=} T_C$

```
# Prove knowledge of (m=3, r=2) in E_alice = (16, 16)
k_m = 1; k_r = 1                                # random nonces
T_R = mod_exp(g, k_r, p)                        # 4
T_C = (mod_exp(g, k_m, p) * mod_exp(pk_a, k_r, p)) % p # 2

e = fs_hash(g, pk_a, R_a, C_a, T_R, T_C)        # 8

s_m = (k_m - e * m) % q                         # 10
s_r = (k_r - e * r_a) % q                       # 7

chk1 = (mod_exp(g, s_r, p) * mod_exp(R_a, e, p)) % p
chk2 = (mod_exp(g, s_m, p) * mod_exp(pk_a, s_r, p) * mod_exp(C_a, e, p)) % p

print(f"Commit:   T_R = {T_R}, T_C = {T_C}")
print(f"Challenge: e = {e}")
print(f"Respond:  s_m = {s_m}, s_r = {s_r}")
print(f"Verify 1:  g^{s_r} * R^e = {chk1:2} == T_R={T_R:2} {'PASS' if chk1 == T_R else 'FAIL'}")
print(f"Verify 2:  g^{s_m} * pk^{s_r} * C^e = {chk2:2} == T_C={T_C:2} {'PASS' if chk2 == T_C else 'FAIL'}")

# Counter-example: Alice claims m'=5 (wrong identity)
m_fake = 5
s_m_fake = (k_m - e * m_fake) % q
chk2_f = (mod_exp(g, s_m_fake, p) * mod_exp(pk_a, s_r, p) * mod_exp(C_a, e, p)) % p
print(f"\nCounter-example (m'={m_fake}):")
print(f" Verify 2: {chk2_f} == T_C={T_C} {'PASS' if chk2_f == T_C else 'FAIL'}")
print(f" The ciphertext encrypts m={m}, not m'={m_fake}.")
print(f" No response value can bridge the gap.")
```

```
Commit:   T_R = 4, T_C = 2
Challenge: e = 8
Respond:  s_m = 10, s_r = 7
Verify 1:  g^{s_r} * R^e = 4 == T_R= 4 PASS
Verify 2:  g^{s_m} * pk^{s_r} * C^e = 2 == T_C= 2 PASS
```

Counter-example (m'=5):  
 Verify 2: 4 == T\_C=2 FAIL  
 The ciphertext encrypts m=3, not m'=5.  
 No response value can bridge the gap.

**How it works:** The prover commits to random nonces  $k_m, k_r$  before seeing the challenge  $e$ . Only someone who knows the real  $m$  and  $r$  can produce responses that satisfy both equations for the specific  $e$  the Fiat-Shamir hash generates. Check 1 confirms knowledge of  $r$ ; Check 2 confirms knowledge of  $m$  and its consistency with the ciphertext. A wrong  $m'$  produces a wrong  $s_m$ , which ripples through the multi-exponentiation in Check 2.

### 14.3 Chaum-Pedersen: Proof of Same Plaintext (Approve)

At **approve** time, Alice re-encrypts her identity for Bob and proves both ciphertexts contain the same  $M$  – without revealing it. Three independent checks verify her key ownership, her knowledge of the new randomness, and the binding constraint that the decrypted values match.

Statement:  $C_a \cdot (R_a^{sk_a})^{-1} = C_b \cdot (pk_b^{r_b})^{-1}$  (same  $M$ )  
 Commit:  $T_1 = g^{k_1}, \quad T_2 = g^{k_2}, \quad T_3 = pk_b^{k_2} \cdot R_a^{-k_1}$   
 Challenge:  $e = H(g, pk_a, pk_b, R_a, C_a, R_b, C_b, T_1, T_2, T_3)$   
 Respond:  $s_1 = k_1 - e \cdot sk_a, \quad s_2 = k_2 - e \cdot r_b \pmod{q}$   
 Verify 1:  $g^{s_1} \cdot pk_a^e \stackrel{?}{=} T_1$  (owns  $pk_a$ )  
 Verify 2:  $g^{s_2} \cdot R_b^e \stackrel{?}{=} T_2$  (knows  $r_b$ )  
 Verify 3:  $pk_b^{s_2} \cdot R_a^{-s_1} \cdot (C_b/C_a)^e \stackrel{?}{=} T_3$  (same  $M$ )

```
# Prove E_alice=(16,16) and E_bob=(13,1) encrypt the same M
k1 = 1; k2 = 1
T1 = mod_exp(g, k1, p) # 4
T2 = mod_exp(g, k2, p) # 4
T3 = (mod_exp(pk_b, k2, p) * mod_exp(R_a, (q - k1) % q, p)) % p # 12

e_cp = fs_hash(g, pk_a, pk_b, R_a, C_a, R_b, C_b, T1, T2, T3) # 4

s1 = (k1 - e_cp * sk_a) % q # 3
s2 = (k2 - e_cp * r_b) % q # 9

v1 = (mod_exp(g, s1, p) * mod_exp(pk_a, e_cp, p)) % p
v2 = (mod_exp(g, s2, p) * mod_exp(R_b, e_cp, p)) % p
cb_ca = (C_b * mod_inv(C_a, p)) % p
v3 = (mod_exp(pk_b, s2, p) * mod_exp(R_a, (q - s1) % q, p)
      * mod_exp(cb_ca, e_cp, p)) % p

print(f"Commit: T1={T1}, T2={T2}, T3={T3}")
print(f"Challenge: e = {e_cp}")
print(f"Respond: s1={s1}, s2={s2}")
print(f"Verify 1 (owns pk_a): {v1:2} == T1={T1:2} {'PASS' if v1 == T1 else 'FAIL'}")
print(f"Verify 2 (knows r_b): {v2:2} == T2={T2:2} {'PASS' if v2 == T2 else 'FAIL'}")
print(f"Verify 3 (same M): {v3:2} == T3={T3:2} {'PASS' if v3 == T3 else 'FAIL'}")

# Counter-example: Alice encrypts M'=g^7=8 instead of M=g^3=18
m_bad = 7; M_bad = mod_exp(g, m_bad, p)
C_b_bad = (M_bad * mod_exp(pk_b, r_b, p)) % p
e_bad = fs_hash(g, pk_a, pk_b, R_a, C_a, R_b, C_b_bad, T1, T2, T3)
s1_b = (k1 - e_bad * sk_a) % q
```

```

s2_b = (k2 - e_bad * r_b) % q
v1_b = (mod_exp(g, s1_b, p) * mod_exp(pk_a, e_bad, p)) % p
v2_b = (mod_exp(g, s2_b, p) * mod_exp(R_b, e_bad, p)) % p
cb_ca_b = (C_b_bad * mod_inv(C_a, p)) % p
v3_b = (mod_exp(pk_b, s2_b, p) * mod_exp(R_a, (q - s1_b) % q, p)
        * mod_exp(cb_ca_b, e_bad, p)) % p

print(f"\nCounter-example (M'=g^{m_bad}={M_bad}, C_b'={C_b_bad}):")
print(f" Verify 1 (owns pk_a): {v1_b:2} == T1={T1:2} {'PASS' if v1_b == T1 else 'FAIL'}")
print(f" Verify 2 (knows r_b): {v2_b:2} == T2={T2:2} {'PASS' if v2_b == T2 else 'FAIL'}")
print(f" Verify 3 (same M): {v3_b:2} == T3={T3:2} {'PASS' if v3_b == T3 else 'FAIL'}")
print(f" Alice owns her key and knows r_b -- checks 1,2 pass.")
print(f" But E_alice encrypts M={M} while E_bob encrypts M'={M_bad}.")

Commit: T1=4, T2=4, T3=12
Challenge: e = 4
Respond: s1=3, s2=9
Verify 1 (owns pk_a): 4 == T1= 4 PASS
Verify 2 (knows r_b): 4 == T2= 4 PASS
Verify 3 (same M): 12 == T3=12 PASS

Counter-example (M'=g^7=8, C_b'=3):
Verify 1 (owns pk_a): 4 == T1= 4 PASS
Verify 2 (knows r_b): 4 == T2= 4 PASS
Verify 3 (same M): 1 == T3=12 FAIL
Alice owns her key and knows r_b -- checks 1,2 pass.
But E_alice encrypts M=18 while E_bob encrypts M'=8.

```

**How it works:** Check 3 is the binding constraint. Expanding the verification equation, it passes only when  $C_a/R_a^{s_{ka}} = C_b/pk_b^{r_b}$  – i.e., decrypting  $E\_alice$  with Alice’s key yields the same  $M$  as de-randomizing  $E\_bob$  with  $r_b$ . A different message breaks this equality, producing a mismatch that no choice of responses can hide, because the commitments were fixed before the challenge was known.